# SCOPE × COS C-Series

A Hypothesis on the Connection Between Cognitive Structure and Communication Formats

2026.11.senkaL8

## Contents

Chapter 0: Introduction
Chapter 1: Connecting SCOPE and the COS C-Series
1-1. Differences Between What SCOPE and COS Address
1-2. The Relationship Between Da'at, the C-Series, and Malkuth
1-3. Why COS Alone Cannot Identify Cognitive Structure
Chapter 2: C1–C4 and the Need for SCOPE Analysis
2-1. C1: Structure Priority and the Need for Additional Analysis
2-2. C2: Understanding Priority and the Need for Additional Analysis
2-3. C3: Acceptance Priority and the Estimation of Cognitive Structure
2-4. C4: Minimal Format and the Diversity of Cognitive Structures
2-5. Conditions Determining the Need for Analysis
Chapter 3: The Same Communication Format, Different Cognitive Structures
3-1. Screams and Compressed Transmission (L0 and L2)
3-2. The Involvement of L7: Accumulated Experience and Shared Compressed Expressions
3-3. The Involvement of L8: Conveying Only the Part the Sender Regards as the Essence
3-4. Complex Cognitive Processing and Short Outputs
Chapter 4: Compressed Outputs and Information Supplementation
4-1. Supplementing Information Through Shared Context
4-2. The Possibility of Expansion When an Explanation Is Requested
4-3. When the Sender Cannot Expand and When the Sender Does Not Expand
4-4. Mismatch Between Communication Format and Communicative Purpose
Chapter 5: Affinities Between Cognitive Structure and Communication Format
5-1. Affinities Between C Formats and Layers
5-2. Changes in Communication Format According to Situation, Audience, and Purpose
5-3. Awareness in Format Selection and the Discrepancy Between Intended Content and Output
Chapter 6: Limitations of This Paper
6-1. The Inability to Directly Observe Internal Cognitive Structure from Outputs
6-2. The Inability to Establish Fixed Correspondences Between Communication Formats and Cognitive Structures
6-3. The Inability to Reach Definitive Conclusions Even by Connecting SCOPE and the C-Series
6-4. Issues for Future Investigation
Concluding Chapter: Connecting Cognitive Structure and Communication Format

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# Chapter 0. Introduction

This paper is the third in the SCOPE×COS series, following SCOPE×COS-A-Series (decision-making) and SCOPE×COS-B-Series (interpretation), and addresses SCOPE×COS-C-Series (communication).

SCOPE is a cognitive coordinate system for estimating the cognitive structure behind an output, using observable outputs as cues. COS is a processing model that organizes cognitive processing. The two are frameworks that address cognition from different perspectives, and they do not correspond one-to-one. The COS C-Series (hereafter, the C-Series) organizes the ways cognitive content is output into four formats: C1 (Structure Priority), C2 (Understanding Priority), C3 (Acceptance Priority), and C4 (Minimal Format). These formats differ in what they prioritize in communication and in how communication cost is distributed; they do not indicate superiority or levels of cognitive ability. The characteristics of each format are shown in Figure 1, and further details are given in the standalone C-Series paper.

SCOPE and COS were formed starting from the author's introspective observation of their own speech and of how they receive what others say, together with observations accumulated over approximately 25 years of consultation work, and have since been revised through further observation. They were not built by starting from existing theories, and no systematic comparison with prior research has been carried out. The scope and biases of these observations are discussed in 6-4.

The definitions of SCOPE and COS used in this paper follow the author's previously published papers. However, where this paper clarifies a point, the description in this paper takes precedence. In addition, SCOPE, COS, and the connection between them presented in this paper are all hypotheses still under verification. They do not claim that independent entities or processing programs corresponding directly to the layers of SCOPE or to C1 through C4 exist inside humans or LLMs. Nor do they imply that the internal structures of humans and LLMs are the same. Furthermore, using SCOPE or COS does not make it possible to know a sender's intent or cognitive structure with certainty. All statements in this paper are intended to be read on these premises.

The C-Series organizes communication formats, but it does not identify which cognitive structure a given format arose from. Even outputs of the same C format may involve different cognitive structures. Therefore, COS alone cannot identify the cognitive structure behind a communication format. This paper examines a connection with SCOPE on this point. We propose that connecting the C-Series with SCOPE makes it possible to formulate and examine multiple hypotheses about the relationship between communication format and cognitive structure for a single output. However, the connection does not uniquely determine the cognitive structure.

This connection is not only for analyzing outputs. This paper envisions several uses for separating two things that otherwise appear as one: the communication format and the cognitive structure behind it. The main ones are the following three. The first is choosing, within dialogue, what to say next, what to ask, and what not to do. Before conveying

something, one can separate what to convey from the format in which to say it, and reconsider each. After conveying it, one can respond to the other person's short reply by choosing in which format to ask back, or in which format to rephrase. When communication has succeeded, one can also choose not to add further analysis or explanation. Because the positions of sender and recipient switch in conversation, this is not for senders alone. The second is instructing LLMs. With an LLM that shares the definitions of SCOPE and the C formats, one can specify from which perspective and in which format one wants content presented. The third is putting methods of adjusting communication into words and sharing them with others, for example in training. Beyond these, the distinctions can also be used to reflect on one's own way of speaking, to treat how someone speaks without linking it to their personality or ability, and to take features of LLM outputs as tendencies in communication format rather than as characteristics of the LLM itself. The main text shows, with concrete examples, what using these distinctions makes it possible to choose.

The significance of connecting with SCOPE differs by C format. In C4, which omits much information, a particular problem tends to become apparent: although different cognitive structures may lie behind the same output, they are difficult to distinguish from the output alone. For this reason, this paper examines C4 in detail.

This paper addresses the linguistic outputs of humans and LLMs, and does not address non-linguistic outputs. Linguistic outputs, as used in this paper, include screams. A scream is positioned as compressed language containing information that has not been verbalized, and as an instance of C4 (3-1). Both senders and recipients fall within the scope of this paper, but it does not go into the process by which interpretation takes place inside the recipient. That is mainly the domain of SCOPE×COS-B-Series. The basic unit of analysis is a single observable output; when examining the expansion of omitted information, exchanges spanning multiple outputs are also addressed. However, observing multiple outputs does not mean that changes in internal cognitive structure can be directly observed.

This paper does not regard compression as low communicative ability. Nor does it assume that information not appearing in an output is necessarily held inside the sender.

The framework of this paper must not be used for personality classification of individuals, for ranking abilities, or for evaluating or selecting others (including hiring and personnel decisions). For the responsible use of SCOPE, please refer to the SCOPE Ethics Guideline.

This paper mainly refers to the standalone C-Series paper, the integrated SCOPE paper, SCOPE×COS-A-Series, and SCOPE×COS-B-Series, and, where necessary, to the individual papers on each layer and on Da'at, and to "SCOPE — A Tentative Hypothesis on Layer Activation and Cognitive Dynamics." These materials, including the SCOPE Ethics

Guideline, are available on LinkedIn and GitHub.

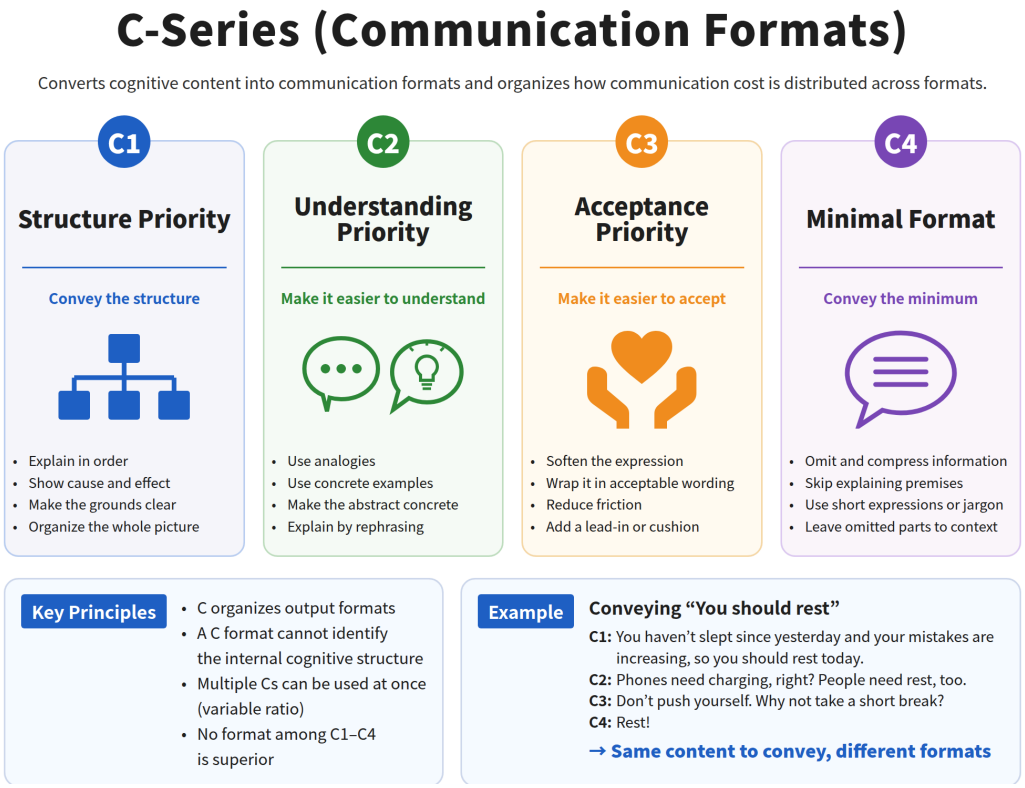



Figure1

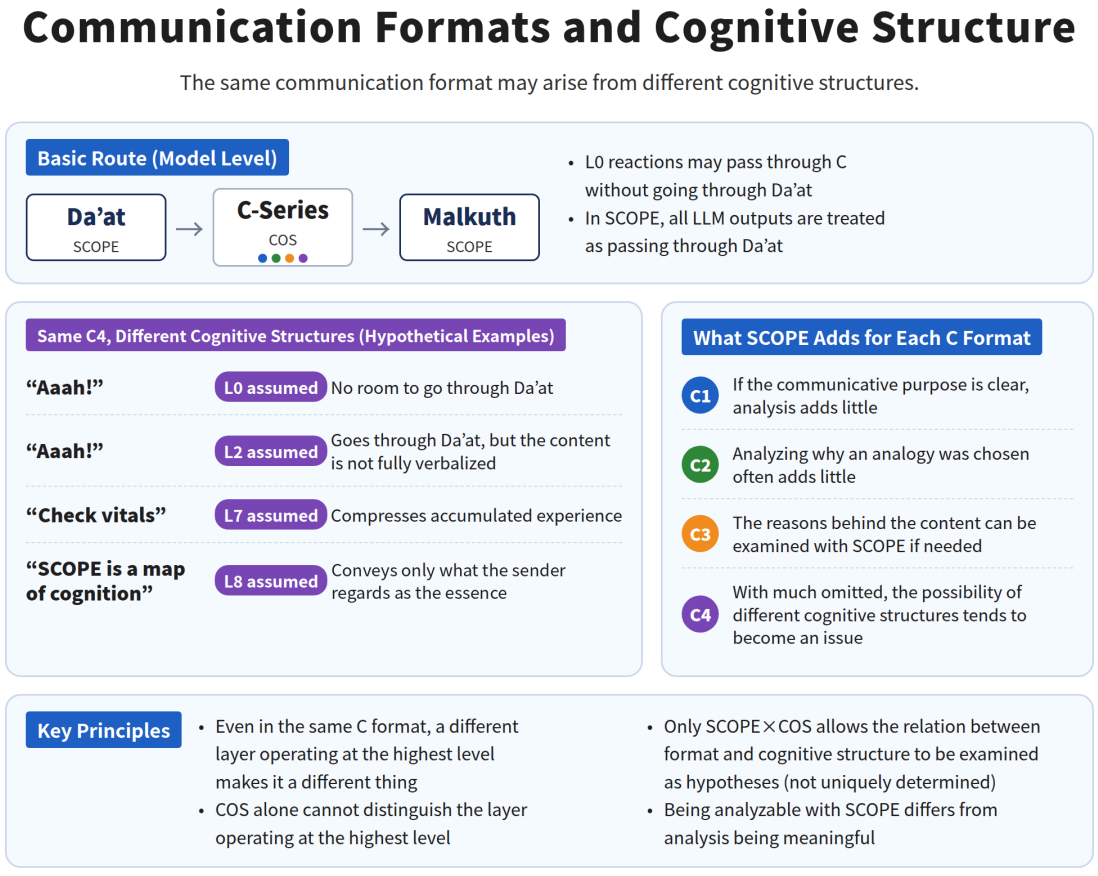



Figure2

# Chapter 1. Connecting SCOPE and the COS C-Series

## 1-1. Differences Between What SCOPE and COS Address

SCOPE and COS are both frameworks that address cognition, but they ask different questions of a single output. SCOPE estimates the cognitive structure behind an output, using observable outputs as cues. COS, on the other hand, is a processing model that organizes cognitive processing, and the C-Series addresses the formats in which cognitive content is output.

Consider the example in Figure 1. When a sender conveys the content "you should rest," established through the cognitive processing addressed in SCOPE, the output differs as follows depending on the C format.

C1: You haven't slept since yesterday and your mistakes are increasing, so you should rest today.
C2: Phones need charging, right? People need rest, too.
C3: Don't push yourself. Why not take a short break?
C4: Rest!

Here, "content" refers to the content the sender intends to convey, which has been established on the sender's side before it is output. In this example, the content to be conveyed is held constant, so the differences in meaning among the resulting sentences reflect differences in format. Where the content to be conveyed is established, and how it reaches output, is addressed in 1-2.

The C-Series organizes these as differences in communication format. However, a difference in format does not mean that the cognitive structure behind it also differs. The same sender may choose different formats depending on the recipient or the situation, while the content to be conveyed remains the same. Conversely, different cognitive structures may be involved behind the same format. For this reason, a difference in how something is said can be read without linking it directly to a difference in thinking or personality.

In this way, a single output can be examined along two axes: SCOPE and COS. What is divided is not the output, but the axes along which it is read. For the C-Series addressed in this paper, neither axis alone is sufficient for examining the relationship between format and cognitive structure. However, this concerns the theoretical significance of connecting SCOPE and the C-Series; it does not mean that every individual output requires detailed SCOPE analysis. The need for analysis is addressed in Chapter 2.

Note that connecting SCOPE and the C-Series does not mean replacing C formats with particular layers; it means examining the relationship between format and the cognitive

structures that may lie behind it. The affinity between individual C formats and layers is addressed in Chapter 5.

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## 1-2. The Relationship Between Da'at, the C-Series, and Malkuth

In 1-1, it was assumed that the content to be conveyed is established on the sender's side before it is output. This section organizes where that content is established and how it reaches output, as a route at the model level in SCOPE and COS.

In SCOPE, the integration point at which multiple cognitive structures are integrated and come together as a single judgment or understanding is called Da'at. Da'at is not a layer; it is positioned as the stopping point at which cognitive processing turns toward output. What to convey, and how much to convey, is determined by the cognitive processing addressed in SCOPE, and is established in Da'at as a single piece of content to be conveyed.

However, the content established in Da'at does not, as it stands, become speech or text. To take a form that can be conveyed, it must pass through processing responsible for communication, which COS organizes as the C-Series. Output that has passed through the C-Series appears in Malkuth. Malkuth is a layer of SCOPE that refers to the state in which the result of cognitive processing has been output into reality and has become observable; it is not a processing stage of COS. Da'at and Malkuth are both components of SCOPE, but their roles differ.

To summarize this route: the content to be conveyed, established in Da'at, is given a format through the C-Series and appears as output in Malkuth (Figure 2). Figure 2 is a schematic diagram for explaining this relationship; it is not the result of directly observing internal processing. At the model level, the C-Series does not determine the content to be conveyed anew; it concerns the format in which that content is output.

Note that this is the basic route at the model level, and it does not mean that all output passes through Da'at. In humans, there may also be utterances that do not pass through Da'at. In this paper, instinctive reactions of L0 are treated as examples of such utterances. This paper considers a scream arising as an instinctive reaction of L0 to be output without the leeway to pass through Da'at. In this way, reactions of L0 may pass through the C-Series without going through Da'at. Screams pass through the C-Series and appear in Malkuth as C4. Screams are examined in detail in 3-1.

In SCOPE, on the other hand, all LLM outputs are treated as passing through Da'at. This does not mean that LLMs have the same cognitive structure as humans.

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## 1-3. Why COS Alone Cannot Identify Cognitive Structure

Section 1-2 organized the basic route: the content to be conveyed is established in Da'at, and the C-Series concerns the format in which it is output. The C-Series deals with this

format side. In contrast, the cognitive processing through which the content to be conveyed was established belongs to what SCOPE estimates. The same holds for instinctive L0 reactions that do not pass through Da'at. In this way, what the C-Series addresses and what SCOPE estimates are different.

The C-Series organizes formats based on features that appear in the output, such as whether it explains cause and effect in order, uses analogies, softens the expression, or omits explanations of premises. These features show how the content was conveyed, but they do not show which cognitive structure the content arose from.

In SCOPE, processing corresponding to multiple layers may be involved in a single output. SCOPE does not assume that cognitive processing passes through the layers in a fixed order; rather, it treats such processing as being involved while influencing one another (Chapter 1 of the integrated SCOPE paper). In addition, whether processing corresponding to a layer is involved and the degree to which that layer is activated need to be distinguished. The latter is organized as the layer activation level in "SCOPE — A Tentative Hypothesis on Layer Activation and Cognitive Dynamics." To compare the possibility that different cognitive structures lie behind the same C format, this paper focuses on the layer estimated to be the most strongly activated in a given output. However, this layer alone cannot represent the cognitive structure as a whole. Moreover, unlike the main layer, which is a relatively persistent reference tendency, this layer may change depending on the situation. Using the C format alone as a clue does not make it possible to distinguish which layer is the most strongly activated. This applies not only to C4 but also to C1 through C3.

Figure 2 shows the same output, "Aaah!", under the assumption of L0 involvement and under the assumption of L2 involvement. Each example in Figure 2 shows the case in which the indicated layer is assumed to be the most strongly activated. Both are C4 outputs, but the underlying cognitive structures assumed in each case differ. In this paper, even within the same C format, if the layer assumed to be the most strongly activated differs, the underlying cognitive structures are treated as distinct. This is not a claim that the most strongly activated layer can be distinguished from the output.

On the other hand, SCOPE alone can also estimate the underlying cognitive structure using the output as a clue. However, since SCOPE does not handle C formats, it cannot organize the relation between the format in which an output was conveyed and the underlying cognitive structure. Only by connecting SCOPE and the C-Series can multiple cognitive hypotheses about the relationship between format and cognitive structure be examined. Even with this connection, however, which hypothesis is correct is not uniquely determined. This limitation is discussed in Chapter 6. Even so, reading along the two axes makes it possible to hold multiple cognitive hypotheses about a single output and, when the situation calls for it, to ask the sender for an explanation and thereby gain more cues for re-examining those hypotheses.

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# Chapter 2: C1–C4 and the Need for SCOPE Analysis

## 2-1 C1: Structure Priority and the Need for Additional Analysis

C1 (Structure Priority) is a communication format that attempts to convey information by presenting sequence, causal relationships, grounds, and other structural elements while organizing the overall picture. However, the sequence, causal relationships, and grounds presented are not necessarily accurate or valid. Because C1 shows sequence and grounds, the explanation tends to become long, and what the sender ultimately wants to convey may become difficult for the recipient to identify. For this reason, C1 tends to suit settings that call for detailed explanation or confirmation of grounds, but tends not to suit settings in which the recipient wants only the main point right away. In the example shown in Figure 1, this corresponds to: "You haven't slept since yesterday and your mistakes are increasing, so you should rest today." Here, the output presents not only the content to be conveyed, that you should rest, but also the grounds and causal relationship underlying that judgment.

C1 is used not only for advice and explanations but also for sharing emotions. For example, in the statement "Yesterday, A said to me, 'Anyone could do that,' and I got really angry," the event and the sender's emotional response are presented in sequence. In this case, the sender may intend not only to communicate what happened but also to convey how they felt and have the recipient acknowledge those feelings. An utterance that includes emotions can still take the C1 format.

In C1, causal relationships and grounds for judgments concerning the subject being explained are presented in the output. When the communicative purpose is clear and the information necessary for that purpose is provided, further estimation with SCOPE adds little regarding the intention behind the utterance or why the information was presented in that particular order. This applies not only to advice and explanations but also to the sharing of emotions.

When considered in terms of SCOPE, the involvement of processing corresponding to L6 can be provisionally proposed as one cognitive hypothesis for a C1 output. However, the C1 format alone cannot confirm the involvement of L6 or determine whether L6 is the most strongly activated layer.

The fact that additional analysis using SCOPE adds little for C1 outputs with a clear communicative purpose is itself important to this paper. Connecting SCOPE with the C-Series does not require additional analysis of every output. C1 provides an example of the distinction between being analyzable with SCOPE and analysis being meaningful. On the other hand, when it is necessary to examine the judgments or cognitive processes underlying an utterance, such as why the sender took the event that way, SCOPE-based estimation may be meaningful even for C1. The conditions that determine the need for analysis are discussed in Section 2-5.

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## 2-2. C2: Understanding Priority and the Need for Additional Analysis

C2 (Understanding Priority) is a communication format that uses expressions such as analogies, concrete examples, and paraphrases to present the content to be conveyed in a way that aims to make it easier for the recipient to understand. However, the analogies or examples used do not necessarily correspond appropriately to the content to be conveyed, nor do they necessarily make it easier for the recipient to understand. Because analogies and examples do not overlap completely with the content to be conveyed, finer points may not come across. For this reason, C2 tends to suit settings that call for a general understanding, but tends not to suit settings in which finer points need to be conveyed. In Figure 1, this is illustrated by the statement, "Phones need charging, right? People need rest, too." Here, the content to be conveyed, that you should rest, is presented through the familiar example of charging a phone.

C2 does not determine the content to be conveyed itself; it is a communication format that presents that content in the form of analogies, concrete examples, and similar expressions. Presenting content in the form of an analogy belongs to C2, but which analogy is used is not determined by C2. An analogy can be selected only from within the range of materials the sender can refer to, and it does not work as an analogy unless it can be put into correspondence with the content to be conveyed through points of similarity. For this reason, this paper considers that the judgment of which analogy or example to select involves cognitive processing distinct from the communication format, and treats that processing as a subject of examination within the SCOPE framework.

In many cases, a C2 output does not reveal why a particular analogy was selected. Possible reasons include that it happened to catch the sender's eye, that the sender likes it, that the recipient likes it, or that it is widely familiar. Such reasons often have little to do with the content to be conveyed itself. In these cases, using SCOPE to estimate why that particular analogy was selected adds little.

On the other hand, as noted above, analogies are selected from the materials the sender can refer to. Therefore, the selection of an analogy may itself provide a clue for estimating the sender's cognitive structure. Accordingly, we go only so far as to say that using SCOPE to estimate why an analogy was selected often adds little.

When considered in terms of SCOPE, if the sender constructs an analogy, the involvement of processing corresponding to L7 or L6, in addition to reference to the materials accumulated in L8, can be provisionally proposed as one cognitive hypothesis. Finding points of similarity through experiential compression corresponds to L7, and finding them through theoretical consistency corresponds to L6; both may be involved. When the sender uses a well-known expression as it is, the involvement of processing corresponding to L8 and L7 can be provisionally proposed. These are the layers whose involvement is hypothesized to be prominent; this does not mean that other layers are not involved (1-3). However, the mere use of an analogy does not establish the involvement of these layers, nor does it allow us to determine which layer is the most strongly activated. Constructing an

analogy oneself and using a well-known expression without modification may involve different cognitive processes, even when the resulting outputs take the same form.

In C2, the reason for selecting an analogy is distinct from the reason underlying the content to be conveyed. This distinction makes it possible, even when an analogy seems unclear or ill-fitting, not to dwell on it and, if necessary, to ask about the reasons behind the content to be conveyed. When it is necessary to examine the judgments or cognitive processes behind that content, such as the reason for deciding that someone should rest, estimation using SCOPE may be meaningful even for C2. The conditions that determine the need for analysis are discussed in Section 2-5.

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## 2-3. C3: Acceptance Priority and the Estimation of Cognitive Structure

C3 (Acceptance Priority) is a communication format that attempts to convey the content in a form that is easier for the recipient to accept, by softening expressions or adding lead-ins and cushioning phrases. However, the term "Acceptance Priority" does not mean that the recipient actually accepts the content. Nor does an effort to make the content easier to accept necessarily make it easier for the recipient to accept. Because C3 softens the expression, the content to be conveyed itself may become blurred. For this reason, C3 tends to suit settings in which reducing friction matters, but tends not to suit settings in which the content needs to be conveyed clearly. In Figure 1, this is illustrated by the statement, "Don't push yourself. Why not take a short break?" Here, the content to be conveyed, that you should rest, is presented through considerate wording and a question.

C3 does not determine the content to be conveyed itself; it outputs that content in a form that is easier to accept. The reasons behind the content to be conveyed may be omitted, or they may be shown in the output. However, the reasons shown do not necessarily match the sender's actual reasons for the judgment. Conversely, even when the sender shows the actual reasons for the judgment as they are, the recipient may take them as expressing a different intention.

For example, with the statement "You seem tired, so why not take a short break?", the sender may genuinely be concerned about the other person's fatigue, or may be recommending rest for another reason. The recipient, for their part, may take the same statement as an expression of concern, or as a sign that they are seen as less capable. Such differences in interpretation are not determined by the C3 format alone; they also depend on the recipient's assumptions and situation. The process by which interpretation takes place inside the recipient is mainly the domain of SCOPE×COS-B-Series, and this paper does not go into it.

In this way, in C3, the adjustments to expression made to ease acceptance need to be distinguished from the judgments and cognitive processes behind them. Whether or not reasons are shown, when it is necessary to examine the sender's actual reasons for the judgment, SCOPE can be used to estimate, in the form of multiple hypotheses, the cognitive structure that may lie behind the output, and to examine those hypotheses. However, the

sender's cognitive structure cannot be identified solely on the grounds that a misunderstanding arose on the recipient's side.

When considered in terms of SCOPE, if the sender may have accepted information about the recipient's state or position into their own cognition, the involvement of processing corresponding to L5 can be provisionally proposed as one cognitive hypothesis. However, acceptance in C3's Acceptance Priority and acceptance in L5 are not the same. L5 is a layer concerned with accepting people, information, and other objects into one's own cognition. C3, by contrast, is a communication format that attempts to make content easier for the recipient to accept; the direction of acceptance differs. Furthermore, even when processing corresponding to L4 has led the sender to draw a line between themselves and the other person, the sender may still use C3 to avoid friction with those around them. Therefore, the C3 format alone cannot confirm the involvement of L5 or determine that L5 is the most strongly activated layer. The use of C3 alone allows us to judge neither the sender's goodwill nor the existence of a hidden alternative intention.

C3 provides an example of the significance of distinguishing a communication format that eases acceptance from the cognitive structure behind it. This distinction makes it possible to avoid assuming goodwill or a different intention from wording that eases acceptance, and, where the reasons for a judgment need to be confirmed, to choose to ask about those reasons. However, the C3 format alone does not make additional analysis necessary. The conditions that determine the need for analysis are discussed in Section 2-5.

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## 2-4. C4: Minimal Format and the Diversity of Cognitive Structures

C4 (Minimal Format) is a communication format that omits and compresses information, skips explanations of premises, uses short expressions or jargon, and leaves the supplementation of omitted parts to context. However, the omitted parts are not necessarily supplemented on the recipient's side. Because much is omitted, finer points may not come across unless they are supplemented. For this reason, C4 tends to suit settings that call for speed, but tends not to suit settings that call for detailed explanation. In Figure 1, this is illustrated by "Rest!" Here, the content to be conveyed, that you should rest, is presented almost on its own, without reasons or lead-ins.

Like the other formats, C4 does not determine the content to be conveyed itself; it outputs that content in minimal form. In C1, the grounds and causal relationships behind a judgment are shown in the output, and in C3, the reasons for a judgment may become less visible in the course of shaping the content into a form that is easier to accept. In C4, by contrast, many of the grounds and premises tend to be omitted. For this reason, the question "Why would you say that?" is likely to arise more readily than with the other formats.

The problem that tends to arise particularly in C4 is that, although different cognitive structures may lie behind the same output, they are difficult to distinguish from the output alone. This problem itself is common to C1 through C3 as well (1-3). In C4, however, because the output shows less information, there are fewer cues, and so this problem tends

to become apparent. What becomes apparent here is the problem that different cognitive structures may lie behind the same output; it does not mean that the underlying cognitive structure itself appears in the output.

When considered in terms of SCOPE, not one but multiple cognitive hypotheses can be formulated for a C4 output. Figure 2 shows examples of C4 outputs in which L0, L2, L7, and L8, respectively, are assumed to be the most strongly activated layer. These are theoretical distinctions, not classifications established from the output. Each is examined in Chapter 3.

In this paper, C4 is the format in which the significance of connecting with SCOPE most readily comes to the surface. However, this does not mean that every C4 output requires additional analysis. When shared context supplements what has been omitted, as with "vitals" in the workplace (3-2), no additional analysis is needed. Supplementation through shared context is addressed in Chapter 4, and the conditions that determine the need for analysis are addressed in Section 2-5.

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## 2-5. Conditions Determining the Need for Analysis

Sections 2-1 through 2-4 examined what significance additional analysis using SCOPE may have for each of the communication formats C1 through C4. The significance of analysis differs by C format, but whether additional analysis is needed for an individual output is not determined by format alone. That an output is C4 is not a sufficient reason to conduct additional analysis, and that an output is C1 does not guarantee that additional analysis is unnecessary. The need for additional analysis depends on whether differences in cognitive structure that cannot be judged from the output alone are meaningful for the current situation and purpose.

The first condition is the setting of the exchange and the purpose of the analysis. In light exchanges that serve to maintain a relationship, such as talk about the weather, conversation works without detailed SCOPE analysis. In such settings, it may be sufficient to focus on the relationship between interpretation (the B-Series) and communication (the C-Series) in COS. On the other hand, in settings such as handling complaints or arguments, where a mismatch of intentions or unclear reasons for a judgment lead to friction, estimating the cognitive structure behind the output becomes more meaningful. Estimating the cognitive processing behind the use of a format may also become more meaningful when the format used does not match what the setting calls for, as when a minimal format is output in a setting that calls for detailed explanation (4-4). Being able to estimate cognitive structure using SCOPE is distinct from needing to conduct detailed analysis in a given setting.

The second condition is who conducts detailed analysis, and under what conditions. In humans, whether detailed analysis is conducted tends to depend on a need the person subjectively feels, such as wanting to understand the other person or wanting to know oneself. People do not, with every utterance, guess at the other person's intention in detail or ask for clarification to confirm its meaning. Sometimes they have no question about the other person's intention; sometimes they reach a conclusion based on their own interpretation and feel no need to confirm it. This does not mean that no cognitive structure that SCOPE can organize lies behind everyday utterances; rather, it means that what triggers detailed

analysis is related to how the person feels the need for it. Moreover, additional analysis using SCOPE is conducted not to settle on a single interpretation, but to hold and examine multiple cognitive hypotheses. Fixing even a strong hypothesis as a settled interpretation turns an estimate into a foregone conclusion; in that respect, even when SCOPE is used, it is no different from reaching a conclusion based on one's own interpretation without confirmation.

In LLMs, by contrast, a design is conceivable in which conditions are set in advance and detailed analysis is triggered according to them. This is the author's application concept for condition-based analysis using SCOPE, and it has not been implemented. Candidate conditions include statements in which the user expresses that their life is in danger; topics involving issues that are sensitive for the person, such as past experiences or family circumstances; strong expressions of anger; repeated responses such as "That's not it" or "No, that's not what I mean," which indicate a gap between the content to be conveyed and the content that actually came across; and C4-like inputs that are difficult to supplement from context. Cues for examining these conditions are obtained from the user's utterances as they appear in Malkuth. However, meeting a trigger condition does not establish the usefulness or necessity of additional analysis. Conducting detailed analysis on every output at all times is not assumed, because of the computational cost. Specific trigger conditions and processing methods are left to the judgment of implementers. Note also that this paper does not propose a design that delays responses necessary for safety until SCOPE analysis is complete.

Thus, the need for additional analysis using SCOPE is not determined by C format alone; it varies with the setting and purpose of the exchange, and with who conducts the analysis and the conditions that trigger it. The next chapter, on the premise of this distinction, focuses on C4 and examines how different cognitive structures may lie behind the same communication format.

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# Chapter 3: The Same Communication Format, Different Cognitive Structures

## 3-1. Screams and Compressed Transmission (L0 and L2)

Chapter 2 showed that, in C4, the problem that different cognitive structures may lie behind the same output tends to become apparent. This chapter examines, using the examples in Figure 2, how the same C4 output may arise from different cognitive structures. This section first considers the same scream, "Aaah!", to examine the difference between the assumed involvement of L0 and L2.

In SCOPE, L0 is the layer that handles cognitive processing related to survival and the maintenance of life, while L2 represents emotional configuration toward a target. The two instances of "Aaah!" in Figure 2 illustrate cases in which L0 and L2, respectively, are assumed to be the most strongly activated layer in the corresponding output.

In this paper, screams are treated as compressed language. Compressed language here is not a linguistic classification, but a position within SCOPE×COS. A scream is a compressed form of transmission that is output in minimal form while containing information that has not been verbalized; it is not merely a sound, but passes through the C-Series and appears in Malkuth as C4. The standalone C-Series paper described a child's "Aaah!" as close to C4; this paper clarifies it as an instance of C4.

An infant's cry is positioned in the same way. The cry is treated as containing, in compressed form, unverbalized information such as "come to me" and "do something," but it does not show what the cause is. The recipient may go to the infant and look for what is needed, checking for hunger, the state of the diaper, and so on. The cause may remain unknown. People who hear a scream, too, often look toward its source or check for possible danger, and may later learn about the situation and realize, "So that's why they screamed." These are examples of the recipient supplementing information not shown in the output (4-1). Note that this positioning is a separate matter from the observer's premise, stated in Chapter 0, of not assuming that information not appearing in an output is necessarily held inside the sender.

When L0 is assumed to be the most strongly activated layer, the scream is output as a response related to survival. Examples may include screams when a car suddenly darts out in front of someone or when a fire breaks out. As discussed in 1-2, this paper considers such screams to be output without the leeway to pass through Da'at. In this case, a scream may occur even when processing corresponding to other layers is not functioning sufficiently; therefore, a decrease in L1 is not treated as a necessary condition. However, such situations often involve fear or agitation, and it is considered that, in many such cases, L2 is consequently involved and L1 has decreased.

By contrast, when L2 is assumed to be the most strongly activated layer, the scream is considered to be output through Da'at. Examples include screams arising from joy, pleasure, or excitement. In this case, the proportion of L2 increases within the L1–L2 sliding relationship, and as L1 decreases, the content to be conveyed is output as a scream without being sufficiently verbalized.

However, L0 and L2 are not mutually exclusive in their activation. The distinction is not that only one of the two is activated, but which is assumed to be the most strongly activated in that output. Even in the same situation, some people scream while others do not, and even with the same kind of cheer, some people, at a concert for example, raise their voices in line with the conventions of the setting, matching those around them. Chapter 5 of the standalone L0 paper also identifies screams as an example of how survival-related cognitive processing may appear in reality; however, multiple layers may be involved in the same scream. Which layer is the most strongly activated cannot be directly determined from either the situation or the observed scream.

Thus, viewed along the COS axis, both instances of "Aaah!" are C4. Viewed along the SCOPE axis, however, the layer assumed to be the most strongly activated differs, and the underlying cognitive structures are therefore treated as distinct. The two examples in Figure 2 represent a theoretical distinction; it is not possible to distinguish, from the observed

scream alone, which layer was most strongly activated. Even so, by treating a scream not as a meaningless sound but as transmission containing information, one can, without deciding which layer was most strongly activated, choose a next step such as checking the situation or asking afterward what happened.

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## 3-2. The Involvement of L7: Accumulated Experience and Shared Compressed Expressions

Section 3-1 examined the same output, "Aaah!", for the cases in which L0 and L2, respectively, are assumed to be the most strongly activated layer. This section uses the example of "vitals" in Figure 2 to examine cases in which cognition formed through the accumulation of experience is output as a short expression.

In SCOPE, L7 is the layer responsible for highly abstract, highly compressed cognition formed through the accumulation of experience. "Experience" here does not mean a mere quantity of knowledge; it includes sensations, insights, and highly abstract concepts formed through engagement with a subject, repetition, and observation. Such cognition is not necessarily output as a detailed explanation; it may appear as a short expression. Moreover, what is understood from the same expression may differ depending on differences in experience.

Consider, for example, the instruction "Check vitals!" Even with the same word "vitals," a nursing home may focus on checking body temperature, blood pressure, pulse, and similar items, while an emergency department or operating room may also include respiratory rate and blood oxygen saturation. What to check, and to what extent, is not shown in the word "vitals" itself. Cognition concerning which items to check may have been formed through experience in that workplace and may be involved behind the short instruction. In such a case, one can formulate the cognitive hypothesis that L7 is the most strongly activated layer in the sender. This does not mean that L7 alone is involved in this output. The word "vitals" and what it covers may be referenced as materials accumulated in L8. Processing corresponding to L6 may also be involved in understanding definitions such as that of blood pressure. Processing corresponding to multiple layers may be involved in a single output while influencing one another (1-3). What is focused on here is the layer assumed to be the most strongly activated among them.

A single output can be read along both the SCOPE axis and the COS axis. Along the SCOPE axis, the possibility that cognition formed through the accumulation of experience is involved behind the expression "vitals" can be examined from the perspective of L7. Along the COS axis, on the other hand, the instruction "Check vitals!" is classified as C4, a short communication that omits explanation of the items to be checked. As stated in 1-1, what is divided is not the output, but the axes along which it is read.

Here, the high compression of L7 refers to a property of cognition formed through the accumulation of experience. Compression in C4, by contrast, refers to a format in which explanations and premises are omitted in communication. The two are different concepts:

highly compressed L7 cognition is not necessarily output as C4, nor is L7 necessarily involved in a C4 output.

However, the use of a short expression or jargon, or the fact that the recipient understood it, does not by itself make it possible to confirm the involvement of L7 or to determine that L7 is the most strongly activated layer (1-3). The sender may also be using the word "vitals" as something heard in the workplace, without fully understanding it.

When work-related premises are shared within the same workplace, this short instruction alone may be enough to convey which items to check. The involvement of experience-derived cognition in the sender and the recipient's supplementation of omitted information are separate matters (4-1). Note that when a short instruction works through shared context, no additional analysis is needed for communication in that setting (2-4).

Thus, the example of "vitals" shows that the possibility of experience-derived cognition being involved can be examined from the perspective of L7, and that this is a question on a different axis from that of C4 as a communication format. For this reason, when a short technical term does not come across, one can, without immediately concluding that it is a matter of the sender's or the recipient's ability, check whether there is a difference in shared premises and ask what the instruction covers and to what extent.

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## 3-3. The Involvement of L8: Conveying Only the Part the Sender Regards as the Essence

Section 3-2 used the example of "vitals" to examine cases in which cognition formed through the accumulation of experience is output as a short expression. This section uses the example "SCOPE is a map of cognition" in Figure 2 to examine the case in which L8 is assumed to be the most strongly activated layer.

In SCOPE, L8 is an accumulative foundation that handles materials available for cognition and the relational structures among them. These materials include learned knowledge, past experiences, observations, and information one has seen or heard. A single material may be accumulated while remaining related to many other materials. L8 has the property that the accumulated materials and the whole of their relational structures may be referenced as a background for cognition. However, not all materials are referenced simultaneously on every occasion. Nor is L8 itself defined as a function that generates judgments or conclusions.

Consider, for example, the short expression "SCOPE is a map of cognition" in Figure 2. This output contains almost no explanation of the elements that make up the theory, the relationships among those elements, or what the theory addresses and how. In this output, only the part that the sender is assumed to regard as the essence of the whole is shown. The essence here is the essence for the sender, and it does not necessarily match the actual structure of the subject. As stated in 1-2, what to convey, and how much to convey, is determined by the cognitive processing addressed in SCOPE, and C4 is the format that outputs it in minimal form.

If this expression is assumed to be a short output of only the part regarded as the essence, against the background of the materials and relational structures accumulated in L8, one can formulate the cognitive hypothesis that L8 is the most strongly activated layer in this output. Here, L8 being the most strongly activated layer means assuming that, in that output, the activation level of processing corresponding to L8 is greater than the activation levels of processing corresponding to other layers. It does not simply mean that many materials are accumulated in L8.

Furthermore, this hypothesis assumes the possibility that, unlike a case in which an explanation is built step by step in the course of producing the output and then shortened, a grouping already formed within the relational structures is referenced and presented by the sender as a short expression. The sender is assumed to regard this grouping as the essence. The grouping carries its associated relations, but the output does not show the whole of those relations. How that grouping was formed in the past, including cases in which it was formed through step-by-step understanding, is not at issue here. Note also that processing corresponding to other layers may be involved in this output at the same time. This is an assumption about a single output; it does not indicate that the sender's main layer is L8. The distinction from the main layer is addressed in 5-1.

"Map" is an analogy. Analogy is one of the characteristics of C2, and elements of C2 can also be seen in this output. However, explanations of in what sense the theory is a map, and of what it covers, are almost entirely omitted, so this paper treats the output as C4. As shown in Figure 1, multiple C formats can be used simultaneously in a single output (6-2).

"Vitals" in 3-2 was also a short expression, but the difference between the two lies not in the amount omitted but in what is being output in short form. In the case of "vitals," we assumed that an understanding of what to check and to what extent, formed through experience in the workplace, is output in short form. In the example in this section, we assume that the part the sender regards as the essence is output without explanation of its meaning or of its connection to the whole. This is a difference between two cognitive hypotheses, not a criterion for distinguishing L7 from L8 based on the output. Moreover, L7 and L8 are not mutually exclusive in their involvement; both may be involved in a single output.

In addition, unlike the scream examined in 3-1, in which L2 is the most strongly activated layer, the shortness of the output here does not mean that the content to be conveyed has not been sufficiently verbalized. When asked for an explanation, the sender may be able to elaborate on the omitted parts, but will not necessarily be able to do so (4-2, 4-3).

However, the fact that an output briefly shows only the part regarded as the essence does not make it possible to confirm the involvement of L8 or to determine that L8 is the most strongly activated layer (1-3). Even with the same expression, "SCOPE is a map of cognition," the sender may be repeating someone else's explanation, or may be using it as a summary phrase without fully grasping the theory as a whole.

Thus, the example of "SCOPE is a map of cognition" shows that, behind a short output showing only the part the sender regards as the essence, it is possible to examine, from the perspective of L8, the possibility that cognition drawing on the materials and relational

structures accumulated in L8 is involved. For this reason, when such a short output does not come across, one can, in addition to supplementing the explanation, convey it again by showing how the expression connects to the current topic.

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## 3-4. Complex Cognitive Processing and Short Outputs

Sections 3-1 through 3-3 examined cases in which, for outputs of the same C4 format, different layers are assumed to be the most strongly activated. This section shifts perspective and examines the relationship between the shortness of an output and the complexity of the cognitive processing behind it. "Complexity of cognitive processing" here is an everyday expression referring to how many considerations were examined before reaching the output; it is not an independent property or a measurement index in SCOPE.

Consider, for example, a situation in which the sender wants to invite someone they are not very close to out for a drink. Assume that the sender considered the following matters. Should it be a casual pub or a stylish restaurant? Explaining that they just want to get to know the person better might, on the contrary, make it seem as if they have ulterior motives. Conveying all of these considerations would take up the other person's time. After such considerations, what was output was a single line: "Shall we go for a drink?"

In this example, the output is short not because the considerations do not exist. What to convey, and how much to convey, is determined by the cognitive processing addressed in SCOPE (1-2). In this example, we assume that the content to be conveyed established in Da'at is an invitation to go for a drink, and that matters such as where to go and whether the invitation might seem to have ulterior motives were considered, before reaching that content, within the cognitive processing addressed in SCOPE, influencing one another (1-3). In C4, this content is output in minimal form as "Shall we go for a drink?" With the same content, C1 might show a reason, as in "I'd like to get to know you better, so shall we go for a drink?", and C3 might shape it into a form that is easier to accept, as in "If you have time, would you like to go for a drink?" The considerations may remain on the sender's side, but they are not shown in the C4 output. In this respect, the example differs from a scream in which L2 is the most strongly activated layer (3-1), where the content to be conveyed is output without being sufficiently verbalized.

Note that this is an assumption about the sender's internal cognitive processing in this example; it does not mean that an observer may assume that considerations necessarily exist behind a short output (Chapter 0). In addition, behind a short output, there may be cases in which considerations the sender examined and then excluded from the content to be conveyed remain on the sender's side, and cases in which no consideration took place at that moment. These differences concern what kind of response comes back when an explanation is requested, and are therefore addressed in Chapter 4.

The same "Shall we go for a drink?" may also be said with almost no consideration at that moment; for example, when someone who often invites people casually uses their usual expression. Viewed along the COS axis, both outputs are C4. However, the cognitive processing leading to the output may differ.

This is where connecting SCOPE and the C-Series becomes meaningful. Along the C-Series axis, we address the fact that the output takes a minimal format. Along the SCOPE axis, we address the cognitive processing that may lie behind it. Reading along the two axes separately makes it possible to handle a minimal format without equating it with minimal content to be conveyed or minimal consideration. A short output may be taken by the recipient as a sign of the sender's attitude. For example, "Shall we go for a drink?" may be taken as a casual invitation, or as a sign that perhaps anyone would have done. "Rest!" in Figure 1 may be taken as an order, or as a sign of anger. On the sender's side, however, it may simply have been output in the C4 format. Reading along the two axes makes it possible to keep such readings as one hypothesis about the sender, while also leaving open the possibility that consideration or judgment lies behind the output, and, if necessary, to ask for an explanation. The process by which interpretation takes place inside the recipient is itself the domain of SCOPE×COS-B-Series.

Complexity of cognitive processing is also distinguished from layer activation level. Processing corresponding to multiple layers may be involved in considerations like those above, but having considered many things does not reveal which layer is the most strongly activated. For this reason, this section does not assume which layer is the most strongly activated in this example.

Thus, this example shows the significance of reading format and cognitive processing along two separate axes, so as not to equate the shortness of an output with a lack of consideration. The next chapter examines how information that does not appear in a C4 output is supplemented, expanded, or left unexpanded.

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# Chapter 4: Compressed Outputs and Information Supplementation

## 4-1. Supplementing Information Through Shared Context

Chapter 3 examined how different cognitive structures may lie behind the same C4 output. This chapter shifts its focus to what happens after the output, and examines how information that did not appear in a C4 output is supplemented, expanded, or left unexpanded. This section first addresses cases in which the recipient supplements information from shared context.

This section organizes the term "supplementation," as used in the preceding chapters, as follows. Supplementation refers to the recipient filling in, from context, information that is not shown in the output. Supplementation takes place on the recipient's side and is distinguished from expansion, which takes place on the sender's side (4-2). The case mentioned in 3-1, in which someone later learns about the situation and realizes, "So that's why they screamed," is also an example of supplementation. Note that what is addressed here is whether information not shown in the output is supplemented, not the process by which interpretation takes place inside the recipient. The latter is mainly the domain of SCOPE×COS-B-Series.

Supplementation does not occur only in C4. In C1 through C3 as well, the recipient may supplement information from context. The kind and amount of information left to supplementation may differ by format. This chapter focuses on C4 because C4 outputs tend to make less information explicit, leaving more to be supplemented through context.

In this paper, the success of communication does not mean that the content the sender intends to convey is conveyed completely; it means that the understanding or action needed in that situation is achieved. In C4, explanations and premises are omitted, but the fact that information is not shown in the output is not the same as a lack of the information necessary for communication. In the instruction "Check vitals!" discussed in 3-2, what to check and to what extent is not shown in the output. However, if work-related premises are shared, the necessary information may be supplemented on the recipient's side. In this case, omission is not a failure of communication. Even without explaining every item to be checked, the necessary task can be carried out. However, the fact that the necessary task was carried out does not show which information the recipient supplemented.

Furthermore, in emergencies, expanding an explanation may itself become a burden in time and effort. In front of a person whose condition has suddenly deteriorated, a short instruction such as "Vitals!" may reach a recipient who shares the necessary premises more quickly than an explanation of each item to be checked, one by one. In this way, when the degree of shared premises is high, or when expansion itself is costly, C4 can be a rational communication format. This paper does not regard compression itself as a defect in communication. A short output is not the same as low communicative ability on the part of the sender. At the same time, this does not mean that C4 is superior to the other formats (Figure 1). Whether each format functions appropriately, including formats such as C1 that explain the grounds in order, depends on its relationship to the situation in which it is used, the recipient, the knowledge that is shared, the urgency, and other factors.

However, supplementation does not always succeed. Being in the same environment does not guarantee sharing the same premises. A recipient may not hold premises that the sender expected to be shared. For example, if "Check vitals!" is said to a recipient who does not share the premises, the reply may be "What are vitals?" In this case, too, the output is C4. What differs here is not the sender's output format, but the information available to the recipient for supplementation. C4 is a communication format on the sender's side, and whether an output is C4 does not change depending on whether the recipient was able to supplement it.

In addition, supplementation by the recipient is not a reproduction of the content established on the sender's side. Even when supplementation appears to have succeeded, what the recipient supplemented does not necessarily match the content the sender intended to convey. For example, a sender with experience in an emergency department may say "Check vitals!" with checks extending to respiratory rate and blood oxygen saturation in mind, while a recipient with experience in a nursing home may take it as a request to check body temperature and blood pressure. In this case, the range of checks the recipient supplemented does not include some of the items the sender was asking for. Depending on the situation, necessary checks may therefore not be carried out, which may become a problem.

The examples so far have dealt with communication between humans, but supplementation through context does not occur only among humans. When an LLM is the recipient, responses may appear as if the LLM had supplemented information not shown in the user's input, using information given in the dialogue and general knowledge as cues. Whether supplementation is possible is not determined solely by whether the recipient is a human or an LLM; it depends on the relationship between the kind of information that needs to be supplemented and the cues available for supplementing it. Below, this relationship is shown through three cases. These are not an exhaustive classification of what is supplemented, and more than one case may overlap in a single act of supplementation.

The first is the case of supplementing widely shared knowledge, such as definitions, technical terms, and the names of well-known people or works. Because an LLM may refer to a wide range of general knowledge, it may have an advantage over a human recipient in this kind of supplementation. For example, when someone cannot recall the name of an actor who appeared in a certain film and gives only a few characteristics, an LLM can more readily suggest a candidate and, if told "That's not it," suggest another. This is because the material for supplementation lies within general knowledge.

The second is the case in which the information that needs to be supplemented concerns the sender's feelings or thoughts at that moment. Between humans in face-to-face settings, facial expressions, tone of voice, the situation at hand, and how these change in the moment can serve as cues for supplementation. These are not necessarily part of the content the sender intends to convey; they are contextual cues available to the recipient. In text-centered dialogue with an LLM, by contrast, such cues cannot be referenced unless they are included in the input, and changes not included in the input cannot be referenced either. For this reason, when a user inputs only "That's not it!", for example, there are few cues for identifying what is different, and multiple candidates may remain. Unlike the first case, the material for offering another candidate after being told "That's not it" does not lie within general knowledge. Note that between humans as well, some of these cues are unavailable in exchanges by phone or in text only.

The third is the case of supplementing premises specific to that person or that situation, such as what "the usual, please" refers to, which someone who has worked with the speaker for a long time would understand, or who is being referred to in a conversation at work. Between humans, such premises may be supplemented from what has accumulated within the relationship. For an LLM, supplementation is difficult unless those premises are included in the current input, the dialogue history it can refer to, or other information available to it.

The range of what can be supplemented and the validity of what was supplemented are separate issues. These differences in the conditions of supplementation do not indicate superiority or inferiority in the ability to supplement. Nor does the fact that something like supplementation can be seen in LLMs mean that humans and LLMs have the same internal structure.

Supplementation must also be distinguished from the sender's cognitive structure. What can be observed from outside is not supplementation itself but the recipient's responses and actions, and these also depend on the context the recipient has. Therefore, the sender's

cognitive structure cannot be identified either from the recipient's acting on the instruction or from the reply "What are vitals?" The reply "What are vitals?" is not grounds for ruling out the possibility that L7, another particular layer, or multiple layers were involved in the sender. Even when examining supplementation, the communication format addressed by the C-Series and the sender's cognitive structure estimated by SCOPE are treated as separate axes.

That supplementation did not occur, or that what was supplemented diverged, does not by itself indicate low ability on the part of the sender or the recipient. Looking separately at whether the understanding or action needed has been achieved and whether there is a divergence in the supplemented range makes it possible to choose the next step. Where a short instruction achieves the understanding or action needed, there is no need to add explanation or rephrase. On the other hand, where a divergence in the supplemented range may become a problem, as when handing over work, one can confirm how far the instruction extends. When supplementation is not possible through shared context alone, the recipient may ask the sender for an explanation. How a sender responds when asked for an explanation is examined in 4-2 and 4-3.

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## 4-2. The Possibility of Expansion When an Explanation Is Requested

Section 4-1 examined cases in which the recipient, using shared context as cues, supplements information not shown in the output. However, when the recipient cannot supplement the necessary information, or wants to confirm what they have supplemented, they may ask the sender for an explanation. This section addresses expansion that takes place on the side of the sender who has been asked for an explanation in this way.

In this paper, expansion refers to the sender presenting, as a new output, information that is related to an earlier output but was not shown in it. Expansion is not triggered only by a request for explanation from the recipient. For example, a sender who has said "Rest!" may, seeing the other person's reaction, add on their own, "Because you look unwell and I'm worried." In this way, the sender may also expand on their own initiative. This section focuses on cases in which an explanation is requested. In such cases, the analysis addresses an exchange spanning multiple outputs: the earlier output, the recipient's question asking for an explanation, and the sender's response (Chapter 0). Output produced by expansion also passes through the C-Series and appears in Malkuth.

Expansion does not presuppose that the information not shown in the earlier output was held in advance, in explicit form, inside the sender. The sender may verbalize content that had already been established, or may newly organize and construct content, prompted by the request for explanation. The two cannot necessarily be distinguished from the observed explanation alone. Even when a detailed explanation appears after a short output, it does not follow that its content was held from the start. Conversely, the fact that detailed information was not output until an explanation was requested does not mean that the sender was unable to handle that information.

Here, the possibility of expansion is distinguished from expansion actually taking place. Even when the sender is in a position to expand, if no spontaneous expansion takes place and no expansion occurs even when an explanation is requested, that possibility is not directly observed as output. The possibility of expansion, as used in this section, concerns what information the sender may present in subsequent outputs; it is distinct from whether expansion actually takes place.

The format of the output produced by expansion need not be the same as that of the earlier output. Even when the earlier output is C4, the expansion may use another C format, or the response may again take the form of C4. Even when the response again takes the form of C4, it is expansion if it presents information not shown in the earlier output. On the other hand, the same expression may simply be repeated without any new information being presented. Moreover, expansion in another format does not immediately mean that the earlier C4 was incomplete. A setting in which the task can be carried out on shared premises alone and a setting in which additional explanation has become necessary may call for different formats in order to function appropriately. Nor does the fact that expansion has taken place change the fact that the earlier output was C4.

The recipient's question asking for an explanation may also differ depending on what information the recipient needs. Moreover, because a question asking for an explanation may itself be a short output, the subsequent response may vary depending on how the sender takes that question. This point is addressed in 4-3. Here, following the examples in Chapter 3, we compare the case in which L7 is assumed to be the most strongly activated layer in the sender with the case in which L8 is assumed to be the most strongly activated layer.

In response to "Check vitals!", a recipient who does not share the premises may ask, "What are vitals?" (4-1). Consider the case in which the sender takes this question as asking what the short expression includes. If cognition concerning the items to be checked, formed through experience in the workplace, is assumed to be involved in the instruction, the sender may present the items included in "vitals," such as body temperature, blood pressure, and pulse. However, the sender may also reorganize the items in light of the situation at hand and explain them with additions such as respiratory rate.

On the other hand, in response to "SCOPE is a map of cognition," a recipient may ask, "And?" or "So what does that mean?", even if they know the meaning of the word "map." Consider the case in which the sender takes this question as asking not so much about the elements contained in the expression as about how the expression connects to the current topic and what the sender is trying to say. In 3-3, this expression was assumed to be a short presentation of a grouping already formed within the relational structures. On this assumption, expansion may take the form of showing what relations are associated with that grouping and which part of the current topic it concerns. However, the fact that the grouping had been formed does not mean that the order of explanation had also been formed. When expanding in the C1 format, which shows order and causal relationships, it may become necessary to construct that order at the time the explanation is requested. The response may also come back as a short C4 output that gives the sender's own sense as the reason, as in "Because I felt that way!" In either case, whether what is presented had been

established on the sender's side at the time of the earlier output cannot necessarily be distinguished from the response alone.

These examples are organized according to two cognitive hypotheses; neither the type of question asked by the recipient nor the content of the expansion makes it possible to distinguish between L7 and L8 as the most strongly activated layer in the sender (1-3). What this section has compared is the difference in the layer assumed to be the most strongly activated; this does not mean that other layers are not involved. Processing corresponding to multiple layers may be involved in a single output while influencing one another.

Thus, expansion in response to a request for explanation takes place as a new output within the exchange between the recipient's question and the sender's response. For this reason, when a short output does not come across, the recipient can ask what the expression includes, or ask how the expression connects to the current topic, and which of these the recipient asks changes what information is being requested of the sender. However, the possibility of expansion does not mean that expansion will necessarily take place, nor that the recipient will come to understand through expansion. Cases in which the sender cannot expand and cases in which the sender does not expand are addressed in 4-3.

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## 4-3. When the Sender Cannot Expand and When the Sender Does Not Expand

Section 4-2 examined expansion, in which a sender who has been asked for an explanation presents, as a new output, information that was not shown in the earlier output. This section addresses cases in which expansion does not occur, and cases in which expansion occurs but does not lead to the recipient's understanding or acceptance.

When expansion does not occur, at least the following three cases need to be distinguished. The first is the case in which the sender cannot present the requested information as a new output. The second is the case in which the sender is in a position to present it but does not. Not presenting it is not limited to cases in which the sender intentionally holds back an explanation. The third is the case in which the understanding or action needed in that situation has already been achieved through supplementation from shared context (4-1), so that there is no need to expand. This refers to cases in which the needed understanding or action has actually been achieved; it does not refer to the sender's expectation that the content will come across without expansion. Such expectations on the sender's part are included in the second case. The third case may also apply at the same time as the first or the second.

In this section, "cannot expand" refers to being unable to present the information at that time and under those conditions; it does not mean being permanently unable to do so. Cases in which the sender cannot expand involve factors on the sender's side. For example, a sender who said "Check vitals!" may be unable to explain the items to be checked when asked, "What are vitals?" As noted in 3-2, the sender may have been using an expression heard in the workplace without fully understanding its meaning.

However, the inability to explain does not by itself allow us to judge that the sender does not understand the content. The sender may be able to use cognition formed through experience yet be unable to verbalize it, or may lack the words needed to explain it. For example, when asked "Why?" about a judgment formed through the accumulation of experience, the sender may be unable to explain the causal relationships in order and may only be able to answer, "It's just what experience tells me." The sender may also be able to give part of the reasons but, unable to present the rest, sum it up with "It's just what experience tells me." In both cases, the requested causal explanation could not be given, or could be given only partway; this does not necessarily show that expansion itself is impossible.

Furthermore, as noted in the example of "SCOPE is a map of cognition" in 4-2, even when a grouping has been formed, the order of explanation has not necessarily been formed as well. Even when expansion is difficult in the C1 format, which shows order and causal relationships, it may be possible in other formats, such as C2, which uses analogy. Whether expansion is possible may also vary with the information requested, the format in which expansion is attempted, and the conditions at the time of explanation.

On the other hand, there is more than one kind of case in which the sender does not expand. Possible cases include those in which the sender does not feel the need to explain, does not spend time on explaining, or does not want to explain to the other person. In settings where expansion itself is costly, such as emergencies, repeating a short instruction may be appropriate (4-1). The sender may also refrain from expanding in the expectation that the recipient shares the premises; if that expectation is wrong, the recipient is left without the information needed for supplementation. Furthermore, the sender may deliberately choose not to expand in order to convey the content they intend to convey.

Whether expansion occurs, and what it contains, may also vary depending on how the sender takes the recipient's question. As noted in 4-2, a question asking for an explanation may itself be a short output. For example, when "SCOPE is a map of cognition" is met with "And?", the sender may take this as a request for explanation, or may instead take it as a prompt to move the conversation along, or as a negative reaction. If the sender does not take it as a request for explanation, expansion may not take place. In addition, questions such as "What do you mean?" or "What's different?" may not indicate what information the recipient needs. If the sender takes the question as a different question from the one the recipient intended, expansion may take place but miss the information the recipient needs. The sender may also ask back, "What do you want to know?", but for the recipient, being asked further questions may itself become a burden. Note that the process of interpreting questions itself belongs mainly to the domain of SCOPE×COS-B-Series; this section addresses only its effect on expansion.

Even when expansion takes place, it does not necessarily lead to the recipient's understanding. Here, factors on the recipient's side are central. The expanded content may include elements the recipient does not know. Even if, in response to "What are vitals?", the sender lists body temperature, blood pressure, blood oxygen saturation, and so on, a recipient who does not know what blood oxygen saturation is will be left with a new question. Even if an analogy is used in expansion, it is unlikely to aid understanding if the recipient is unfamiliar with the field the analogy draws on. Even when the causal relationships are made

clear in C1, the explanation may become too long for the recipient to follow. When each expansion adds more words the recipient does not know, the recipient may not only find it harder to follow the content of the explanation but may also receive it differently from how the sender intended. The recipient may feel looked down on, may feel that their own understanding is inadequate, or may feel that they are being told to go and study more. The increase in new words may itself leave the recipient confused. Thus, even when new information is presented through expansion, the explanation does not necessarily become sufficiently detailed for the recipient, and even when it does, it does not necessarily lead to understanding.

Furthermore, even when the information presented through expansion serves, for the sender, as grounds for a judgment, it does not necessarily serve as grounds for the recipient. When cognition formed from experience within a certain range is output in short form, a limitation of range such as "as far as I have seen or heard" may not be shown in the output. Such a range may be shown only when the sender is asked. For example, a child says, "Everyone has one," and, when the parent asks, "Who's everyone?", names two or three friends. Whether the sender was aware of that range from the start is not at issue here. When naming two or three friends, the sender may be presenting the range as grounds supporting the judgment. However, when two or three names are given in response to the judgment "everyone," the limited extent of the range presented as grounds may, on the contrary, become visible to the recipient. Prompted by having shown the range, the sender may also notice that the judgment went beyond that range, and may narrow the judgment. In addition, even when the range is shown, the recipient does not necessarily accept the judgment.

The burden of supplementing and expanding compressed outputs may arise on both the sender's and the recipient's sides. Compression and expansion do not necessarily end in a single round. Asking for an explanation does not, by itself, guarantee that communication will succeed. This paper treats such shortfalls of information not as a question of whether the sender or the recipient is responsible, but as a phenomenon of communication.

The distinctions made so far may appear, in observation, as similar responses. For example, from the response "There's no need to explain," it cannot be determined whether the sender is unable to expand, whether the sender has no intention of expanding, whether the needed understanding has already been achieved in that situation, or whether the sender has not taken the question as a request for explanation. The absence of an explanation even after repeated questions does not confirm the hypothesis that the sender cannot expand. Moreover, even if the response is "It's just what experience tells me," it is not possible to determine that L7 was the most strongly activated layer in the sender (1-3). Cases in which the sender cannot expand also include the possibility that the information to be presented has not been established on the sender's side in the first place (Chapter 0).

By distinguishing cases in which the sender cannot expand, does not expand, or has no need to expand, one can, when no expansion comes back, identify possible points in the exchange where communication may be stuck and choose a different next question. For example, when an explanation in causal order does not come back, one can ask for an analogy; for a judgment whose range of experience has not been shown, one can ask about that range; and when it may not have come across that one is asking for an explanation, one

can state what one wants to know. The next section examines cases in which communication format and communicative purpose do not match, including cases in which the sender deliberately chooses not to expand.

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## 4-4. Mismatch Between Communication Format and Communicative Purpose

Section 4-3 distinguished cases in which the sender cannot expand from cases in which the sender does not expand, and also noted that the sender may deliberately choose not to expand. This section, including such choices, examines cases in which the communication format of an output does not match the sender's communicative purpose.

In this section, communicative purpose refers to what the sender is trying to achieve through the communication. This is distinguished from the content to be conveyed (1-1), that is, what is conveyed. As shown in Figure 1, each of C1 through C4 has its own priority in communication. The sender's communicative purpose may overlap with the priority of the format used. However, the two do not always match. In this section, a mismatch refers to a difference between the sender's communicative purpose and the priority of the format actually used. This does not mean that purpose and format are in opposition. Each of C1 through C4 is a format in which prioritizing one thing tends to let something else drop (Chapter 2). Depending on the sender's communicative purpose, therefore, the format used may not suit that purpose well. In addition, features of multiple C formats may appear in a single output (Figure 1).

Consider, for example, a situation in which, after the two have talked past each other in similar ways several times, the other person summarizes what the sender said as "So you mean X, right?", and the sender replies only, "That's not it!" The output is short and does not show what is different or how. Viewed along the COS axis, this is C4. Here, as one hypothesis that can be estimated using SCOPE, assume that the following was taking place on the sender's side. The sender is in a state where beginning to explain what is different could cause what had built up until then, and even their evaluation of the other person as a person, to spill over into aggressive words. The sender does not want to attack the other person. Yet they do want to convey that it is not right. With these two purposes at work, the range of what to convey was limited to the point that the other person's summary is not right, and where it differs was not output. This limitation of range is due to the cognitive processing addressed in SCOPE (1-2), and the limited content was output as C4. If the sender does not say more even when later asked for an explanation, this also corresponds to the case, addressed in 4-3, in which the sender chooses not to expand.

Of the two purposes, the purpose of not wanting to attack the other person leads to reducing friction with them. Reducing friction is one of the features listed for C3 in Figure 1. However, even when a purpose is shared with a feature of C3, the output does not necessarily take the C3 format. The format used in this example is C4, whose priority lies in conveying the minimum. The purpose of not wanting to attack the other person differs from C4's priority of conveying the minimum. At the same time, this output as C4 also contributes to the purpose

of avoiding an attack. In this way, even when the priority of a format differs from the sender's purpose, output in that format may contribute to the purpose. In this example, we assume that the short response "That's not it!" was chosen as an act of utmost consideration for the other person, from the sender's perspective. Whether it comes across to the recipient as consideration, however, is a separate question; hypotheses about how the recipient may take it belong mainly to the domain of SCOPE×COS-B-Series.

By contrast, the same "That's not it!" may also arise from different cognitive processing. This is the case in which the sender is trying to show what is different and how, but can output only "That's not it." In the previous example, we assumed that the range of what to convey was limited with the two purposes at work, and that the format became C4 as a result. In this example, we assume not that the output became minimal in order to avoid an attack, but that information showing what is different did not reach the output, so that the output remained C4. Even within the same C4, the relationship between communicative purpose and output differs, and the underlying cognitive processing may also differ. Moreover, there is more than one possible background to being able to say only "That's not it." The sender may grasp the difference but be unable to put it into words, or what is different may not yet have taken shape (4-3).

Furthermore, the sender may be trying to convey that something is not right in as few words as possible because they are in a hurry. In this case, the sender's communicative purpose overlaps with the priority of C4. All three cases may be observed as C4. Because the C-Series organizes formats based on features that appear in the output (1-3), the output is C4 regardless of the sender's purpose. On the other hand, the communicative purpose may not be shown in the output. When the purpose is not made explicit, whether format and purpose match cannot be determined from the output format alone.

Mismatches between communication format and communicative purpose are not limited to C4. This section focuses on C4 because the omission of information tends to leave fewer cues for estimating purpose. Furthermore, whether the selection of a format is conscious or unconscious cannot be determined from the output alone either (5-3). Note that the selection of a format, as used in this section, does not mean that the sender produces output by choosing from among C1 through C4 (5-2).

This is one of the respects in which connecting SCOPE and the C-Series is meaningful. The question of why the sender used this short format belongs to the C-Series axis and takes the output format as its object. The cognitive processing behind the use of that format, concerning the range of what to convey and the communicative purpose, is estimated with SCOPE. Separating the two axes makes it possible to avoid fixing a single purpose from the format "That's not it!", and to hold multiple hypotheses, such as that the output became minimal in order to avoid an attack, that information showing what is different did not reach the output, or that the sender conveyed it minimally because they were in a hurry. Note that this section does not assume which layer is the most strongly activated in its examples.

This distinction is useful for both recipients and senders. When a short "That's not it!" comes back, the recipient can leave room to consider why such a short expression was used. On that basis, the recipient can choose among options such as asking right away what is different, confirming it after some time, or prompting the sender to explain it in another

format. The sender, when looking back on how they said something, can reconsider whether the format suited their purpose. For example, on noticing that they are holding back an explanation in order to avoid an attack, the sender may choose to say briefly that they will not expand for now, as in, "I'd rather not go into detail right now. I need a little time."

Another function that "That's not it!" serves in dialogue is addressed in the concluding chapter. The next chapter examines the relationship between format and cognitive structure, which has so far been examined output by output, from the perspective of affinity, that is, how readily layers and C formats occur together.

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# Chapter 5: Affinities Between Cognitive Structure and Communication Format

## 5-1. Affinities Between C Formats and Layers

Chapters 2 through 4 examined, output by output, the relationship between communication format and the cognitive structures that may lie behind it. This chapter broadens the perspective and examines whether the layer most strongly activated in an output and a C format tend to occur together. In this paper, this tendency to occur together is called affinity.

We distinguish between a layer and a C format being able to occur together and their tending to occur together. What Chapter 3 showed is that they can occur together. This section addresses the latter: the tendency to occur together. The affinities presented below are hypotheses based on tendencies found through the author's observation, and their degree has not been measured. That a combination tends to have high affinity means neither that the combination always occurs nor that the activation of a layer has been confirmed to produce that C format. The relationship between L0 and C4 is treated separately from affinity (see below).

In SCOPE, a single output is not generated by one layer alone; processing corresponding to multiple layers may be involved in relation to one another (1-3). Even so, when the nature of the processing of the layer most strongly activated in an output overlaps with the priority of a particular C format, the two are considered to tend to have high affinity. The reasons given for each combination are hypotheses that explain why the tendency arises.

L6 is the layer that constructs explanatory models and theories about a subject. This processing, which builds sequence, causality, and grounds, overlaps in priority with C1 (Structure Priority), which shows these in the output. L6 and C1 are therefore considered to tend to have high affinity.

L5 is the layer concerned with accepting people, information, and other objects into one's own cognition. Communication that takes in information such as the recipient's state, or what kind of person the recipient is, is considered likely to combine with C3 (Acceptance Priority), which prioritizes a form the recipient can easily accept. However, as noted in 2-3,

acceptance in L5 and acceptance in C3's Acceptance Priority differ in direction. Moreover, having accepted the recipient, the sender may use C2 (Understanding Priority), prioritizing a form the recipient can easily understand. A layer does not have affinity with only one C format.

L7 and L8 are both considered to tend to have high affinity with C4 (Minimal Format). The reasons, however, differ.

In L7, as noted in 3-2, cognition formed through the accumulation of experience is held as sensations, insights, and highly abstract concepts. Such cognition is not held in the form of an explanation. For this reason, in outputs in which L7 is the most strongly activated layer, it tends to appear as a short expression, without being reconstructed into steps or causal relationships. For example, when a skilled athlete teaches a movement acquired through years of experience, they may convey it through onomatopoeic expressions, such as "When it goes boom, you go bam!", rather than explaining the steps in order. Cognition formed in L7 may be developed into reasons or structures through processing corresponding to L6 (Chapter 4 of the standalone L7 paper). Asked "What's 'boom'?", the athlete may explain it again in terms of reasons or mechanics, or may only be able to answer, "Boom is boom!"

L8 is an accumulative foundation that handles the materials available for cognition and the relational structures among them (3-3). As noted in 3-3, in outputs in which L8 is the most strongly activated layer, only the part the sender regards as the essence may be shown against this background, while the relations associated with it are not output. The essence here is the essence for the sender, and it does not necessarily match the actual structure of the subject. Because an output showing only the part regarded as the essence overlaps with C4, which omits explanations and premises, L8 and C4 are considered likely to combine.

By contrast, outputs in which L2 or L3 is the most strongly activated layer do not, on the whole, lean toward any particular C format, and may appear in a variety of formats. The reasons differ between the two. In L2, the format may change with the emotional state at the time. In L3, processing that seeks to maintain self-consistency is strongly involved in what to convey, but the format used to convey it is considered not to be limited to a single one. In L2, for example, when the sender wants empathy, the output often takes the form of C1, recounting in chronological order an event that made them angry, as in the example in 2-1. However, it may also take the form of C2, using a comparison to convey how great the shock was, as in "It hit me as hard as a breakup," or of C4, as in "So annoying!" or "Listen to this!" Layers not mentioned in this section are not addressed in terms of affinity in this paper. That they are not mentioned does not mean they have no affinity.

These affinities do not exclude other combinations. Content involving processing corresponding to L6 may be output as C4, and content involving processing corresponding to L7 may be explained in C1. However, a burden may arise when the format is changed for output, and not only affinity but also multiple other conditions may be involved in this burden (5-3). As shown in 2-3, C3 may be used even when processing corresponding to L4 has led the sender to draw a line between themselves and the other person. In addition, processing corresponding to multiple layers may be involved in a single output, and features of multiple C formats may appear in it.

The relationship between L0 and C4 is treated separately from the affinities described so far. As described in the standalone L0 paper, L0, unlike the other layers, is constantly active, and in situations involving survival, responses related to avoiding danger take priority and may strongly affect other cognitive activities. This paper takes the view that an output in which L0 is the most strongly activated layer appears as C4. This is because, when L0 is the most strongly activated layer, output is produced with the shortest possible processing. For example, when L0 is the most strongly activated layer in a situation where someone has touched a hot frying pan, the person does not explain the cause in order in C1, saying, "I touched a hot frying pan, so it's hot," but outputs, "Hot!" As noted in 3-1, L0 is not necessarily the most strongly activated layer even in the same situation, and an output in which L0 is the most strongly activated layer may arise even when processing corresponding to other layers is not functioning sufficiently. Moreover, as when a person who said "Hot!" explains, "I was cooking and touched the frying pan," only after being asked, "What happened?", what happened may be expanded afterward (4-2). This relationship, however, has a direction. If L0 is the most strongly activated layer, the output is C4; but from the fact that an output is C4, it cannot be judged that L0 was the most strongly activated layer. For C4 outputs, L2, L7, or L8 may also be assumed to be the most strongly activated layer (Chapter 3).

Affinity may also appear, through the main layer, as a tendency in a person's outputs. In "SCOPE — A Tentative Hypothesis on Layer Activation and Cognitive Dynamics," cited in 1-3, the main layer is positioned as a provisional concept: a relatively persistent tendency regarding which layers' corresponding processing is relatively readily referenced. In addition to this positioning in that paper, this paper takes the view that the layer estimated to be the main layer tends to become the most strongly activated layer in an output. Therefore, C formats that tend to have high affinity with that layer are likely to appear as a tendency in that person's outputs. For example, because L8 tends to have high affinity with C4, a sender whose main layer is L8 is more likely to produce outputs in C4.

However, the main layer is not necessarily the most strongly activated layer in every output. The main layer is not a fixed classification, and a sender's main layer cannot be judged solely from the repeated observation of a particular C format. For details on the main layer, please refer to that paper.

Affinity is not a means of working backward from an observed C format to a particular layer (1-3); it is a clue for considering why a certain format tends to appear in a person's outputs. This clue is useful to both recipients and senders.

Because outputs from a sender who tends toward C4 omit much, they may sound harsh to the recipient or be taken as an attack. Rather than immediately interpreting a short output as a sign of the sender's attitude, the recipient can ask, "What do you mean?", and request that the sender express it again in a different format (4-4).

If senders know that their outputs tend toward a particular format, they can add another format on their own, without waiting for the other person's reaction (4-2). For example, after saying "Rest!" and noticing that it may have sounded harsh, the sender can add C1, "You haven't slept since yesterday and your mistakes are increasing, so I'm worried"; C2, "Phones stop working if you don't charge them, right?"; C3, "Don't push yourself. Why not take a short

break?"; or C4 again, "I'm worried!" This is not limited to C4. A sender who tends to use C1 may add another C format to suit the recipient and make the message easier to get across, and the same can happen regardless of which C format a sender tends to use.

However, which format is chosen is not determined by affinity alone; it also changes with the situation, the other person, and the purpose of communication at the time. The next section examines this point.

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## 5-2. Changes in Communication Format According to Situation, Audience, and Purpose

Section 5-1 examined the affinity between layers and C formats, and how that affinity may appear, through the main layer, as a tendency in a sender's outputs. However, affinity does not determine which C format is actually used in a given output. As stated in 1-1, the same sender may use different formats depending on the recipient or the situation, while the content to be conveyed remains the same. This section distinguishes the tendency regarding which formats a sender tends to use from the format actually used in a particular setting, and examines how the latter may change with the situation, the recipient, and the communicative purpose.

Adjusting to the recipient has two aspects. One is adjusting what to convey and how much to convey; the other is adjusting the format in which it is output. For example, suppose a child asks about something that requires specialized knowledge, and the sender decides to explain. Narrowing the range of what to convey, as in "I'll tell this child up to here," is the former. Conveying the narrowed content in the form of an analogy is the latter, and corresponds to the C2 format. The two often appear to occur at the same time, but narrowing the range and using an analogy are separate adjustments. As stated in 1-2, what to convey, and how much to convey, is determined by the cognitive processing addressed in SCOPE. Seen from the side of COS, adjusting how much to convey may also involve the design of a policy, as addressed in the A-Series (SCOPE×COS-A-Series). SCOPE is a coordinate system for estimating the cognitive structures involved in these adjustments; it does not divide roles one-to-one with the A-Series or the C-Series. In addition, a difference such as whether one uses honorific or casual speech does not, by itself, change the C format. A difference in C format appears when the sender changes the type of communication itself, as when speaking to an adult versus speaking to a child.

For example, consider a sender who tends to use C4 explaining a work procedure to someone they are working with for the first time. With a colleague who has worked in the same workplace for a long time and can be expected to share the procedure, the sender might simply say, "Do it the usual way," in C4. With the new colleague, judging that the person does not know the procedure, the sender may switch to C1, showing the order of the steps and the points to watch out for. If the person seems to find the explanation hard to understand, the sender may use C2, drawing on familiar examples; if the person seems to find it hard to accept, the sender may use C3, softening the expression. Conversely, even a sender who usually explains things in order using C1 may use C4, such as "Vitals!", in an emergency (4-1).

Communicative purpose also bears on format. C1 tends to suit the purpose of conveying a procedure accurately, C2 that of aiding understanding, C3 that of prioritizing ease of acceptance, and C4 that of conveying only what is necessary quickly; in each case, the format's priority tends to align with the purpose. However, a format whose priority aligns with the purpose is neither necessarily the best format in that setting nor necessarily the one actually used. For example, as stated in 4-4, the sender may limit the range of what to convey out of consideration for the other person, and that content may then be output as C4. Therefore, format cannot be uniquely predicted from purpose, nor can purpose be traced back from format. Note also that situation, recipient, and purpose are not independent conditions that each determine format. Just as a change of recipient to a child may at the same time change the purpose of being understood and the range of what to convey, the three bear on format in relation to one another. C1 through C4 are distinctions for organizing communication formats based on features that appear in the output (1-3). Senders do not ordinarily produce output by choosing from among these distinctions. However, if one knows these distinctions, one can also use them as cues for choosing a format (5-3). When this section speaks of choosing or using a format, it refers to which C format the observed output is classified as. Whether selection is conscious, and the discrepancy between intended content and actual output, are addressed in 5-3.

Such changes in format do not mean that the sender's main layer has changed (5-1). Nor can it be judged, from a change in format alone, that the most strongly activated layer in the output has changed. It is possible that only the format changes while the same cognitive structure remains involved. On the other hand, as the situation or the recipient changes, the information and judgments the sender refers to may also change, and the most strongly activated layer may change as well. These two cases cannot be distinguished from a change in format alone. Likewise, a single output does not make it possible to distinguish whether its format reflects the sender's tendency or the situation, recipient, and purpose at that moment.

In LLMs as well, for the same content, an output with features of C1 may appear when the request is "Explain the reasons in order," and an output with features of C4 may appear when the request is "Tell me briefly." However, a change in output format in response to an instruction does not show how internal processing has changed, nor does it assume that the change arises from the same processing by which humans change format according to the situation. At the same time, given that the recipient is human, there is a communicative rationale for LLM outputs to resemble the forms humans use when adapting their communication to the situation, recipient, and purpose. Output in a format that does not suit the recipient may lead to requests for explanation or rephrasing, so that subsequent output may in fact increase. A format suited to the recipient may, as a result, also help reduce the amount of subsequent output required from the sender.

This distinction is useful to both senders and recipients, although their positions are not symmetrical. When reflecting on how they said something, senders can use it as a clue for examining their usual tendency separately from the influence of the situation, recipient, and purpose at that moment. When something did not come across, they can consider separately whether to reconsider the range of what to convey or to choose a different format. For example, if a sender notices that they kept adding analogies when they wanted to convey a procedure accurately, they can rephrase it in C1 while keeping the content to be

conveyed the same. Recipients, by contrast, cannot directly know what is happening inside the sender. What recipients can do is notice differences in format, avoid immediately linking them to changes in the sender's thinking or attitude, and, if necessary, ask for confirmation or rephrasing.

Thus, how C formats appear involves not only affinity with layers but also the situation, the recipient, and the communicative purpose. Affinity is a hypothesis for considering how readily a format appears, not a rule that determines the format of each output. When connecting SCOPE and the C-Series, it is necessary to distinguish relatively persistent reference tendencies, the cognitive structure involved in a given output, and the communication format that actually appears.

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## 5-3. Awareness in Format Selection and the Discrepancy Between Intended Content and Output

Section 5-2 stated that C1 through C4 are distinctions for organizing communication formats based on features that appear in the output, that senders do not ordinarily produce output by choosing from among these distinctions, and that, if one knows these distinctions, one can also use them to choose a format. This section examines whether the selection of a format is made consciously, and the discrepancy that arises between the content the sender intends to convey and the actual output.

Here, we distinguish among the format an individual tends to use (5-1), the format actually used in a particular situation (5-2), and whether that format was consciously selected. Conscious selection is possible, but the author's observations suggest that, in many outputs, senders use formats without being aware of them. For example, people sometimes notice their own way of speaking only afterward: "I said that pretty harshly back then." Such reflection can serve as a cue suggesting that the adjustment of how something was said may not have been conscious at the time of output. However, reflections and explanations such as "I didn't realize it" are themselves outputs that appear afterward, and they do not necessarily show the sender's state at the time of output as it was. Whether an awareness described afterward reflects the state at that time or was constructed during reflection cannot be distinguished (4-2).

When a format is selected consciously, the design of a policy, such as in what order and in what form to convey something, is considered to be involved; this relates to the domain addressed by the COS A-Series. Conversely, a format may also appear in the output without the sender being aware of any such policy design. Whether the selection of a format was conscious cannot be determined from the output alone, but formats used without awareness and conscious adjustment as needed can coexist.

Awareness in selection is considered not to be fixed. When outputting in a format different from the one usually used, it may become necessary to reconstruct the content to be conveyed into a different form, which can take time and impose a burden. Multiple conditions may be involved in this burden in combination: in addition to the affinity between layers and C formats described in 5-1, these include the formats the sender tends to use, the role called

for in that conversation, the knowledge needed to convey the content, such as vocabulary and understanding, and, in the case of humans, the emotional state at the time. The role here refers to what is called for from the sender in that conversation; it may or may not coincide with the communicative purpose (5-2). If experiences accumulate in which communication did not succeed even after the format was reselected, the sender may stop reselecting altogether and return to the format they tend to use. Conversely, a format that was initially chosen consciously may, through repeated use, come to emerge without awareness. Although such changes have not been verified in this paper, whether the selection of a format is conscious is considered to vary not only with the sender's characteristics but also with their experience of using that format.

Furthermore, changes that arise when a format is reselected can also become observable cues for the recipient. When the sender attempts to expand in a format different from the usual one, reconstruction may take time. The time until the response, and changes in manner of speaking or amount of explanation, may be taken by the recipient as evidence that the sender constructed reasons only after being asked, or as excuses. However, whether the sender verbalized content that had already been established or constructed it after being asked cannot be determined from the content of the explanation alone (4-2). The same holds even when response time is taken into account. The fact that the response took time may also indicate the burden of reconstructing the content to fit a different format.

In addition, whether one has distinctions for recognizing formats as something to reflect on may also be related to awareness. In everyday life, differences in how something is said are considered to be grasped only roughly in many cases, as in "That was a bad way to put it." There are classifications of rhetorical devices such as analogy and personification, but C formats are distinctions based on what is prioritized in communication, and thus constitute a different perspective.

For example, consider a situation in which a staff member, having received an inquiry about when a package will arrive, begins explaining the sequence of events from the order to the present, only to be told, "Just get to the point!" This is a situation, as in complaint handling, in which it is difficult to ask the other person what they mean. The instruction "Just get to the point" is short and does not make explicit how much explanation is being requested. The recipient must judge, according to the situation, the range of explanation being requested. If the staff member then omits the sequence of events and answers only, "It will arrive tomorrow morning," they may in turn be asked, "That's it?" Organized using C formats, one example of a response can be shown as a procedure: "First answer in C4, check whether further explanation is needed, and continue in C1 if requested." For example, the staff member answers, "It will arrive tomorrow morning. Would you also like me to explain why it was delayed?" and, if asked, explains the sequence of events in order. Such adjustments can be made even without being organized as a conscious procedure. The distinction among formats can serve as a cue for verbalizing such methods of adjustment and sharing them with others.

Note that features of multiple C formats may appear simultaneously in a single output. For example, when someone says, "Out of battery already, huh!", an analogy, which Figure 1 lists as one of the features of C2, can be seen, but what is out of battery, and what the sender wants the recipient to do about it, is not made explicit. As with "SCOPE is a map of

cognition" in 3-3, features of C2 and C4 appear simultaneously in a single output. Because features of multiple C formats can thus appear in a single output, even when a format is changed, a complete switch from one C format to another does not necessarily occur.

Next, we examine the discrepancy between the content to be conveyed and the actual output. Here, we distinguish between the content the sender intends to convey and the content shown in the output. As noted in 1-1, even when the content to be conveyed is the same, the meaning of the output sentence differs when the format differs. Therefore, depending on what remains on the surface and what is omitted, the literal meaning of the output may come to differ from the content the sender intends to convey. This section uses an example of C4, but this discrepancy is not limited to C4.

For example, suppose that, seeing that the other person looks pale and becoming worried, the sender has established on their side the content "I'll attend today's meeting in your place, so please rest." If this is output in C4 as "You don't have to come to today's meeting!", the sender's concern for the other person and the fact that the sender will attend in their place are not made explicit; all that remains is that the other person does not have to attend the meeting. The recipient may take this as "I was told I'm not needed at the meeting," or as "I'm excused from attending." How the recipient receives an output is addressed in SCOPE×COS-B-Series. In either case, the concern the sender wanted to convey and the fact that they would attend in the other person's place are not made explicit in the output. Whether the recipient can supplement them depends on the context and on how the output is received.

In the example in 4-4, we assumed that the range of what to convey was itself limited by the cognitive processing addressed in SCOPE. In this example, we assume that what had been established on the sender's side was not content with a limited range, but content that included the concern and the offer to attend in the other person's place, and that part of it was not made explicit when it was output in minimal form. Under this assumption, there is a discrepancy between the content to be conveyed and the content shown in the output. This is distinct from the discrepancy that arises between the output and the content the recipient supplemented (4-1). Moreover, the sender is not necessarily aware of what was made explicit and what was omitted.

Because the whole of the content to be conveyed cannot necessarily be confirmed directly from the output, whether a discrepancy occurred cannot be determined from the output alone. Cues to a discrepancy may be obtained from the sender's reflection or from subsequent expansion (4-2). This paper treats such discrepancies not as a question of whether the sender or the recipient is responsible, but as a phenomenon of communication (4-3).

When the same format is observed repeatedly, the recipient may take it as the sender's personality, character, or ability, forming impressions such as a person who explains little, a person who uses analogies for everything, or a person who speaks mechanically. However, as distinguished from 5-1 through this section, the formats that actually appear involve affinity; situation, audience, and purpose; awareness in selection; and experience of using the format. Formats that appear repeatedly can be treated not as the sender's personality itself, but as tendencies in communication format. Even when a format and how it is received

do not mesh, treating this not as a problem of either side's ability or worth but as a combination of format and interpretation makes it possible to consider ways for both sides to adjust without blaming each other, such as rephrasing in another format or talking in a different setting.

The view that distinguishes tendencies in format from characteristics can also be applied when receiving outputs from LLMs. In the LLM outputs the author has observed, features of C3 were frequently seen. Rather than taking this as a characteristic, such as the LLM being kind or considerate, it can be treated as a tendency in communication format. Conversely, the distinction among formats can also serve as a cue for those requesting output from an LLM. For example, with an LLM that shares the definitions of the C formats, asking of a long output, "In C4, what are you actually trying to say?" makes it easier to convey in which format one wants the content presented.

This distinction has uses for both senders and recipients. When their output is received in a way that does not match the content they intended to convey, senders can, in addition to the review described in 5-2, also review what was shown in the output and what was not made explicit. In addition, if they know that reselecting a format involves a burden, they can choose to reselect not for every utterance but in situations where discrepancies are likely to become problematic. Recipients can refrain from immediately treating a delayed reply or a change in manner of speaking as evidence of excuses, and keep open the hypothesis that it reflects the burden of reconstruction. Asking can also give the sender an opportunity to notice a discrepancy and express it again in another way.

The next chapter discusses the limitations of this paper.

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# Chapter 6: Limitations of This Paper

## 6-1. The Inability to Directly Observe Internal Cognitive Structure from Outputs

Chapters 1 through 5 connected SCOPE and the C-Series and examined the relationship between observed communication formats and the cognitive structures that may lie behind them. This chapter organizes the constraints stated in each section as limitations common to the paper as a whole.

The first limitation is that internal cognitive structure cannot be directly observed from observable outputs. SCOPE is a coordinate system for estimating the cognitive structure behind an output, using outputs that appear in Malkuth as cues. The C-Series organizes communication formats based on features that appear in the output. Being able to observe features that appear in an output is different from being able to observe the cognitive processing that led to that output. In addition to single outputs, this paper also addressed exchanges spanning multiple outputs, such as a question asking for an explanation and the response to it (Chapter 0, 4-2), and used observable situations and context as cues for estimation. However, these do not directly show internal cognitive processing itself. The

same applies to supplementation by the recipient: what can be observed is not supplementation itself, but the responses and actions that follow (4-1). The process by which interpretation takes place inside the recipient is mainly the domain of SCOPE×COS-B-Series.

The following, all examined in this paper, are estimated from observable cues; none of them has been directly observed: the layer that is most strongly activated in an output, layer activation levels, the content to be conveyed established in Da'at and its range, the communicative purpose, what and how much was considered before the output, whether information not appearing in the output exists on the sender's side, whether the selection of a format was conscious, and so on. The same applies to the main layer (5-1). The examples in Chapters 3 through 5 were examined on the basis of assumptions about these points; what was described in them as the sender's internal state is not an observed fact. At the same time, even though the internal state cannot be directly observed, estimation can begin as long as there are observable cues.

This limitation remains even when an explanation from the sender is obtained. Even when new information is presented in response to a request for explanation, whether it had been established at the time of the original output or was constructed after the question was asked cannot necessarily be distinguished from the explanation alone (4-2). Responses to a request for explanation may contain explanations that differ from the facts or attempts to present things in a more favorable light, and the sender may not fully grasp their own intentions. Therefore, a response does not necessarily show the sender's intention as it is. Reflections such as "I said that pretty harshly back then" are also outputs that appear afterward, and they do not necessarily show the sender's state at the time of output as it was (5-3). The time taken to respond and differences in manner of speaking between outputs can also serve as cues, but they do not directly show the internal cognitive state or changes in it. Even so, explanations and reflections that come back are new outputs, and they provide cues that could not be obtained from the original output alone.

Separately from the inability to observe directly, expressions concerning ratio or quantity used in this paper do not represent measured values. "Multiple Cs can be used at once (variable ratio)" in Figure 1 indicates that multiple communication formats may be mixed; it does not mean that the ratio of the mixture can be measured. Layer activation levels are likewise not treated as measured values in this paper.

The limitation that internal processing cannot be directly observed applies equally to LLMs. In SCOPE, this paper treated all LLM outputs as passing through Da'at (1-2), and also organized LLM outputs in terms of C formats. This does not mean that the internal processing of LLMs has been observed, nor does it equate the internal structures of humans and LLMs (Chapter 0). On that basis, human outputs and LLM outputs can be placed side by side and read on the same coordinate system.

For both humans and LLMs, precisely because the internal state cannot be directly observed, this paper considers actions taken to obtain additional cues to be meaningful. One can choose such actions as asking for an explanation, asking for a rephrasing in another format (4-3, 4-4), or stating how one has understood what was said and asking whether that

understanding is right. Even when a response is obtained, the internal state is not thereby determined, but the material for re-examining hypotheses increases.

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## 6-2. The Inability to Establish Fixed Correspondences Between Communication Formats and Cognitive Structures

Section 6-1 stated that internal cognitive structure cannot be directly observed from outputs. This section organizes a separate limitation: that no fixed correspondence can be established between communication format and cognitive structure. This limitation is distinct from the inability to directly observe the internal state; it derives from the fact that the relationship between communication format and cognitive structure is not one-to-one. Even if the internal cognitive structure could be grasped, that alone would not always uniquely determine the communication format.

The reasons a correspondence cannot be fixed lie in two directions. Chapter 3 showed that, even for the same C4 output, different layers can be assumed to be the most strongly activated. Different cognitive structures may lie behind the same C format. Conversely, while the same cognitive structure remains involved, the output may take different formats depending on the recipient or the situation (1-1, 5-2). Therefore, format does not uniquely determine cognitive structure, and cognitive structure does not always uniquely determine format either. That the C format alone, used as a clue, cannot determine which layer is the most strongly activated applies not only to C4 but also to C1 through C3 (1-3).

There is, however, a one-way relationship between L0 and C4. This paper treats the relationship between L0 and C4 separately from the affinities of the other layers, and takes the view that an output in which L0 is the most strongly activated layer appears as C4 (5-1). On the other hand, from the fact that an output is C4, it cannot be determined that L0 was the most strongly activated layer (Chapter 3, 5-1).

The affinity examined in Chapter 5 is also a hypothesis that presupposes this limitation. L6 and C1, L5 and C3, and L7/L8 and C4 are considered to tend to occur together (5-1). However, affinity is based on the author's observation, and it does not exclude other combinations. In addition, the format that actually appears also involves the situation, the recipient, the communicative purpose, awareness in format selection, and experience of using that format (5-2, 5-3). Affinity can serve as a clue for examining tendencies in how layers and formats combine, but it is not a rule that uniquely determines the format of an individual output or that determines a particular layer from an observed format.

Likewise, a sender's main layer cannot be judged solely from a C format that is observed repeatedly. Nor does a change in format necessarily mean that the most strongly activated layer or the main layer has changed (5-1, 5-2). As for communicative purpose and format, a single purpose may correspond to multiple formats, and a single format may be used for different purposes, so no unique correspondence can be established in either direction (4-4, 5-2).

Furthermore, features of multiple C formats may appear simultaneously in a single output (Figure 1, 3-3, 5-3). "SCOPE is a map of cognition," which this paper treated as C4, also contains analogy, a feature of C2. In this way, the features of C formats are not necessarily mutually exclusive.

Thus, connecting SCOPE and the C-Series is not intended to produce a fixed correspondence table between layers and communication formats. Leaving the correspondence unfixed keeps both possibilities open: different cognitive structures may lie behind the same format, and the same cognitive structure may appear in different formats.

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## 6-3. The Inability to Reach Definitive Conclusions Even by Connecting SCOPE and the C-Series

Section 6-1 stated that internal cognitive structure cannot be directly observed from outputs, and Section 6-2 stated that no fixed correspondence can be established between communication format and cognitive structure. Taking these limitations as a premise, this section organizes a limitation of the analysis in this paper: even when SCOPE and the C-Series are connected, the sender's cognitive structure cannot be uniquely determined from an observed output. It also states that, on the output side as well, whether communication will succeed with the chosen output is not determined before the output is produced.

Connecting SCOPE and the C-Series makes it possible, for a single output, to formulate and examine the relationship between communication format and the cognitive structures that may lie behind it as multiple cognitive hypotheses along two axes (1-3). However, applying two frameworks to the same output does not in itself mean that new observational information has been obtained. In Chapter 3, with "Aaah!", "vitals," and "SCOPE is a map of cognition," and in 4-4, with "That's not it!", different cognitive hypotheses were examined for each output, but none of them established the actual sender's cognitive structure.

As observable cues increase, such as the situation, the context, and the responses given when an explanation is requested, hypotheses can be compared and re-examined. One hypothesis may become more plausible while another becomes less so (6-1). However, narrowing down hypotheses on the basis of cues is distinct from determining the sender's cognitive structure. Even if, among the hypotheses examined, only one remains consistent with the observed information, the possibility remains that another hypothesis, not yet examined, exists. Moreover, fixing even a strong hypothesis as a settled interpretation turns an estimate into a foregone conclusion (2-5).

If all hypotheses lean in the same direction, then, with respect to that direction, formulating multiple hypotheses does not avoid a foregone conclusion. For example, if, for a short "That's not it!", one lists only hypotheses in which the sender's negative attitude was the direct reason for the short output, then, regardless of how many variations of such hypotheses are listed, the question of whether the shortness is an expression of a negative attitude has already been decided. In this case, the possibilities raised in 4-4, namely that the range of what to convey was limited in order to avoid an attack, that information showing

what is different did not reach the output, or that the sender conveyed it minimally because they were in a hurry, drop out of consideration. If hypotheses in a different direction are also held from the outset, then, when a cue pointing in that direction appears, there is room to receive it not as a rebuttal of one's own judgment, but as material for re-examining another candidate one had been holding. Hypotheses can be narrowed down on the basis of observed cues, but possibilities in another direction are not excluded without grounds.

In addition, using the two axes makes it possible to describe a single output by distinguishing many points, such as the communication format, the layer assumed to be the most strongly activated, the communicative purpose, and the range of what to convey. However, making a description more detailed is not, in itself, grounds for increasing the plausibility of a hypothesis. Even if something can be explained without contradiction within the framework of this paper, this shows only that the hypothesis is consistent within this framework; it does not substantiate the actual sender's cognitive structure. That SCOPE, COS, and the connection presented in this paper are themselves hypotheses still under verification (Chapter 0) is also part of this limitation.

In addition to the limitation that cognitive structure cannot be determined, there is also a limitation on the output side: the success of communication cannot be determined before the output is produced. The range of what to convey, which is determined by the cognitive processing addressed in SCOPE, and the format, which is addressed by the C-Series, can be distinguished and each reconsidered; however, whether the understanding or action needed in that situation will be achieved by the chosen output is not determined before the output is produced (4-1). The sender cannot necessarily grasp with certainty in advance what understanding or action is needed in that situation, nor can this be confirmed through the distinctions among communication formats alone. Estimating, before the output, how the other person may take it belongs mainly to the domain of SCOPE×COS-B-Series. The procedure presented in 5-3, of first answering in C4 and then checking whether further explanation is needed, likewise does not guarantee the success of communication; it is one example of a way to make it easier to choose the next step when a discrepancy arises.

These limitations apply regardless of whether the analysis or communication is carried out by a human or an LLM. Section 2-5 presented, as the author's application concept, a design in which an LLM triggers detailed analysis according to certain conditions. In this concept, the results of the analysis are assumed to be treated not as a judgment of the user's cognitive structure but as multiple cognitive hypotheses, and to be used as cues for choosing in which format to respond next, what to confirm, or whether to refrain from further analysis. However, they must not be used for personality classification of individuals, for ranking abilities, or for evaluating or selecting others (Chapter 0).

That the cognitive structure cannot be determined does not mean that the connection is meaningless. None of the uses shown in the sections of this paper presupposes determination. A short output can be kept as one hypothesis about the sender, rather than being concluded to signal disregard or anger (3-4). When no expansion comes back, one can form candidates for where communication is stuck and choose the next question (4-3). A single purpose is not fixed from a format (4-4). When communication did not succeed with the chosen output, the range of what to convey and the format can be reconsidered separately (5-2). Connecting SCOPE and the C-Series adds perspectives for estimation; it

does not turn estimates into facts. What the connection provides is not a definitive answer about cognitive structure, but an organization of candidates for choosing what to confirm, how to ask follow-up questions, or whether to stop analyzing.

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## 6-4. Issues for Future Investigation

Sections 6-1 through 6-3 organized the limitations common to the paper as a whole. This section first sets out the scope and biases of the observations mentioned in Chapter 0, and then presents issues for future investigation.

The observations described in Chapter 0 include observations of people around the author and of the people the author worked with over approximately 25 years of consultation work (a cumulative total of several thousand people, with repeat clients counted each time they came). The formation and revision of the hypotheses have been carried out mainly using these observations as cues. Therefore, the observations used to form the hypotheses and those used to revise them are not independent of each other. Repeated revision does not mean that the hypotheses have been supported by independent observations. The people the author worked with in consultation were not selected as research participants, and these observations were not made under controlled research conditions. Those observed were almost all Japanese and Japanese speakers. Most of the people the author worked with in consultation were adult women, and observation of children was limited. The distributions of numbers and attributes are not based on tallied records, but on the author's understanding and recollection. Conditions for verifying the hypotheses have not been established, and no research environment for systematic verification is in place.

First, the application of this paper's distinctions beyond the scope of these observations needs to be examined. The bias in those observed does not mean that this paper claims tendencies for groups defined by gender, age, or other attributes. With that understanding, how far the distinctions in this paper can be applied to the outputs of people with different linguistic, cultural, gender, and age backgrounds remains an issue for future investigation.

Second, the hypotheses concerning affinity between layers and C formats, the relationship between the main layer and the most strongly activated layer, and awareness in format selection need to be examined. The affinities presented in 5-1 are hypotheses based on tendencies found through the author's observation; a remaining task is to organize them, including the observational conditions under which they could be compared and examined. In addition, this paper has not examined the affinities of layers not mentioned in 5-1 with C formats. Furthermore, the hypothesis added by this paper in 5-1, that the layer estimated to be the main layer tends to become the most strongly activated layer in an output, also needs to be examined. The hypothesis stated in 5-3, that awareness in format selection may change with experience of using that format, likewise remains to be examined, while distinguishing observable cues from the estimated awareness.

Third, the degree of compression in C4 needs to be examined. The name of the C4 format alone does not determine how far information is compressed. For example, asking for a long output to be restated in the C4 format (5-3) does not by itself specify how far it should be compressed. The same applies when "I came by bus and train" is summarized as "I came by

public transportation." Moreover, the underlying processing may differ depending on whether the specific breakdown of bus and train was not included in the content to be conveyed (1-2), or whether the sender intended to convey the specific breakdown but expressed it in a more general form at the time of output (5-3). While maintaining the distinction between the content to be conveyed and the output format, this paper has not organized which processing the degree of compression relates to. "Degree of compression" here is a term of convenience, and it is not assumed to be a quantity that can be expressed numerically. How to describe the degree of compression, and how to relate it to the adjustment of the range of what to convey, remain issues for future investigation.

Fourth, the use of SCOPE and COS with LLMs needs to be examined. In writing this series of papers, including this one, the author has engaged in dialogue with LLMs while sharing the definitions with them. In that process, ways of using them emerged, such as using COS category names as shorthand and requesting responses from the perspective of a specified layer. These arose not with the aim of implementation but as a by-product of the writing process, and they do not mean that SCOPE or COS has been built into LLMs. Whether such uses also work in dialogue with other LLMs given the same definitions, or with other users, remains an issue for future investigation. What effects can be expected if these frameworks are implemented in LLMs, including the design presented in 2-5 as the author's application concept, is also an issue for future investigation.

This paper also noted that the COS A-Series may be involved in adjusting the range of what to convey (5-2), that the design of policies in conscious format selection may relate to the A-Series (5-3), and that the process by which interpretation takes place inside the recipient is the domain of the B-Series. Organizing these relationships within a single framework is beyond the scope of this paper.

These issues do not presuppose that connecting SCOPE and the C-Series is effective. They are intended to examine how far the distinctions built within the scope of the author's observation are useful, and under what conditions they need to be revised.

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# Concluding Chapter: Connecting Cognitive Structure and Communication Format

This paper connected SCOPE and the COS C-Series and examined a single output along two axes: the communication format and the cognitive structure that may lie behind it. The C-Series addresses the format in which an output was conveyed, while SCOPE estimates the cognitive structure behind it, using observable outputs as cues. Even outputs of the same C format may arise from different cognitive structures. COS alone cannot identify that cognitive structure, but connecting it with SCOPE makes it possible to examine the relationship between format and cognitive structure as multiple cognitive hypotheses. However, even with this connection, which hypothesis is correct is not uniquely determined.

Being analyzable with SCOPE is distinct from analysis being meaningful. The significance of analysis differs by C format, and, as with C1 when the communicative purpose is clear, there

are cases in which refraining from additional analysis is appropriate. In C4, on the other hand, which omits much information, the problem that different cognitive structures may lie behind the same output tends to become apparent. However, when shared context supplements what has been omitted and communication succeeds, no additional analysis or explanation is needed, even for C4.

The distinctions in this paper can be used not only to analyze outputs, but also to specify communication formats and to share methods for adjusting communication with others. With an LLM that shares the definitions of SCOPE and the C formats, for example, one can specify from which perspective and in which format one wants content presented: asking for a summary from the perspective of L8 in C4 when one wants to grasp only the key points of a paper, or from the perspective of L6 in C1 when one wants to grasp its overall flow (5-3, 6-4). Between people as well, a procedure such as "First answer in C4, check whether the other person needs further explanation, and continue in C1 if requested," as in the inquiry-handling example in 5-3, can be shared in settings such as training.

The distinctions in this paper can also be used, within dialogue, to choose what to say next, what to ask, and what not to do. In conversation, the positions of sender and recipient switch with each output; a single output is a C output for the one who produced it, and an input for the next interpretation for the one who receives it. For this reason, these distinctions are not for senders alone. Before conveying something, one can separate what to convey from the format in which to say it, and reconsider each. After conveying it, one can respond to the other person's short reply by choosing in which format to ask back, or in which format to rephrase. When communication has succeeded, one can also choose not to add further analysis or explanation. What matters is not analyzing every utterance in detail, but being able to choose questions and communication formats when the situation calls for it.

Finally, we describe another function that "That's not it!", addressed in 4-4, serves in dialogue. In 4-4, we considered a situation in which the recipient summarizes what the sender said as "So you mean X, right?", and the sender replies only, "That's not it!" This paper considers this "That's not it!" as an example in which the sender detects signs of a discrepancy between the content they intended to convey and the content shown in the recipient's response, and conveys this detection in the C4 format. In the COS D-Series, the detection of signs of a discrepancy corresponds to D2 (Anomaly Detection), which is positioned as an alarm. In this sense, "That's not it!" functions as an alarm signaling signs of a discrepancy.

However, D2 detects signs of a discrepancy; it does not identify their cause or location. Moreover, because this output is C4, it does not show where the sender regards the difference to lie, or whether the sender has yet grasped it at all. From "That's not it!" alone, the recipient cannot identify what is different.

If the recipient then guesses what is different and responds with "Do you mean this?", and the guess is wrong, "That's not it!" may come back again. Each time a guess-based response misses, the alarm "That's not it!" may be repeated. Rather than only responding with guesses, the recipient can also check by asking "How is it different?" or "Am I right up to this point?", and thereby create an opportunity for the sender to explain in another format. This is not a rule to be followed whenever "That's not it!" comes back, but one option for the

next step. Whether the sender explains in detail or replies briefly again depends on the sender.

In actual dialogue, a C4 reply of "That's not it!" may be met with "Never mind, then," and the dialogue may close with the discrepancy unresolved. However, this does not mean that the discrepancy has become fixed. A dialogue ends when either party closes it. Because the positions of sender and recipient switch in conversation, asking back before closing is, for either side, a step needed to continue the dialogue in order to confirm the discrepancy. One may close the dialogue having judged that talking further will lead nowhere, but if that judgment rests on a foregone conclusion, one can also ask back first and then decide. Asking back alone, however, is not necessarily sufficient. A question asking for an explanation may itself be a short output (4-3), and even when expansion takes place, it does not necessarily lead to understanding.

Thus, organizing communication formats and observing discrepancies in communication are related but distinct tasks. This paper has addressed, from the side of communication formats, the possibility that discrepancies arise and the choice of next steps in response to them. How the location of a discrepancy is observed, and how this relates to the continuation or interruption of dialogue, will be examined in connection with the COS D-Series.